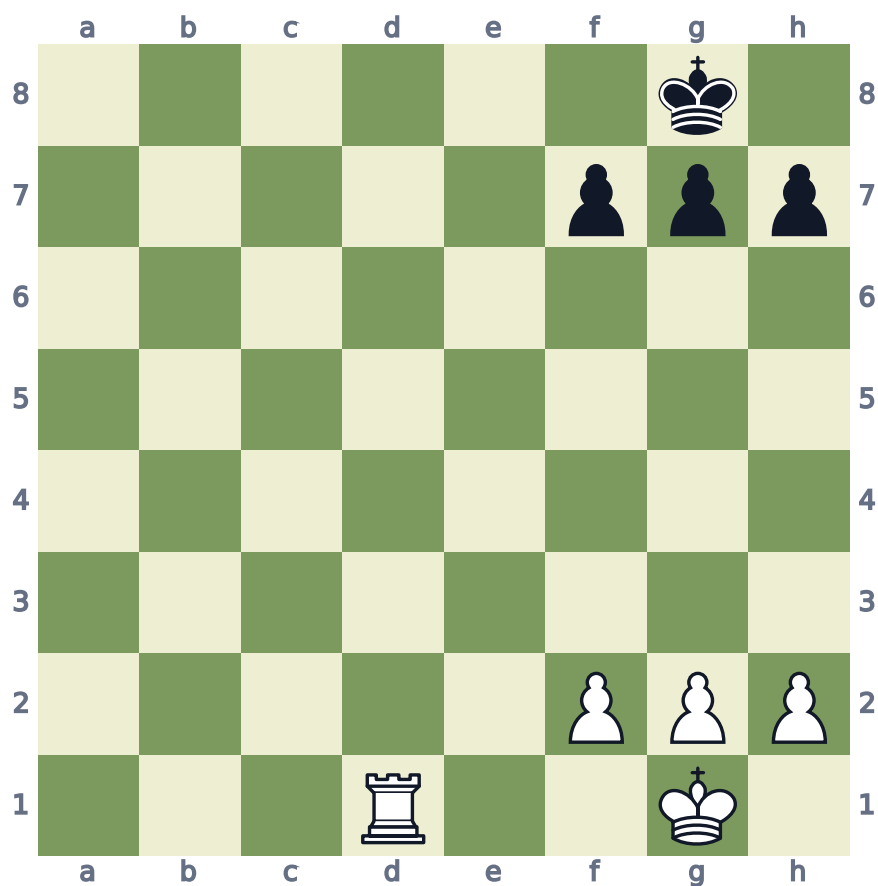


Calculation Test Laboratory

Candidate trees, forcing lines, endpoints, and scoring



Practice calculation as a measurable process rather than a puzzle guess: candidates, best defense, complete variations, endpoint evaluation, and error diagnosis.

24 practice units | separated solutions | scoring | review plan

ORIENTATION

How to get real improvement from this course

PURPOSE

Practice calculation as a measurable process rather than a puzzle guess: candidates, best defense, complete variations, endpoint evaluation, and error diagnosis.

TRAINING RULES

- Use a written tree: candidate, opponent's best reply, continuation, endpoint.
- Return mentally to the root position before starting a different candidate.
- Do not stop a line immediately after winning material; check recaptures and intermediate moves.
- Classify every miss as candidate, visualization, defense, or evaluation failure.
-

COURSE SIZE

This PDF contains 24 practice units. Problems appear before their solutions whenever practical. Review misses after 1, 3, 7, 14, and 30 days.

SCORING

What a complete answer contains

RECOGNITION

Name the tactical or strategic feature without assuming that the familiar pattern works.

CANDIDATES

List serious forcing and improving choices before analyzing deeply.

BEST RESISTANCE

Calculate the opponent's least cooperative legal response.

ENDPOINT

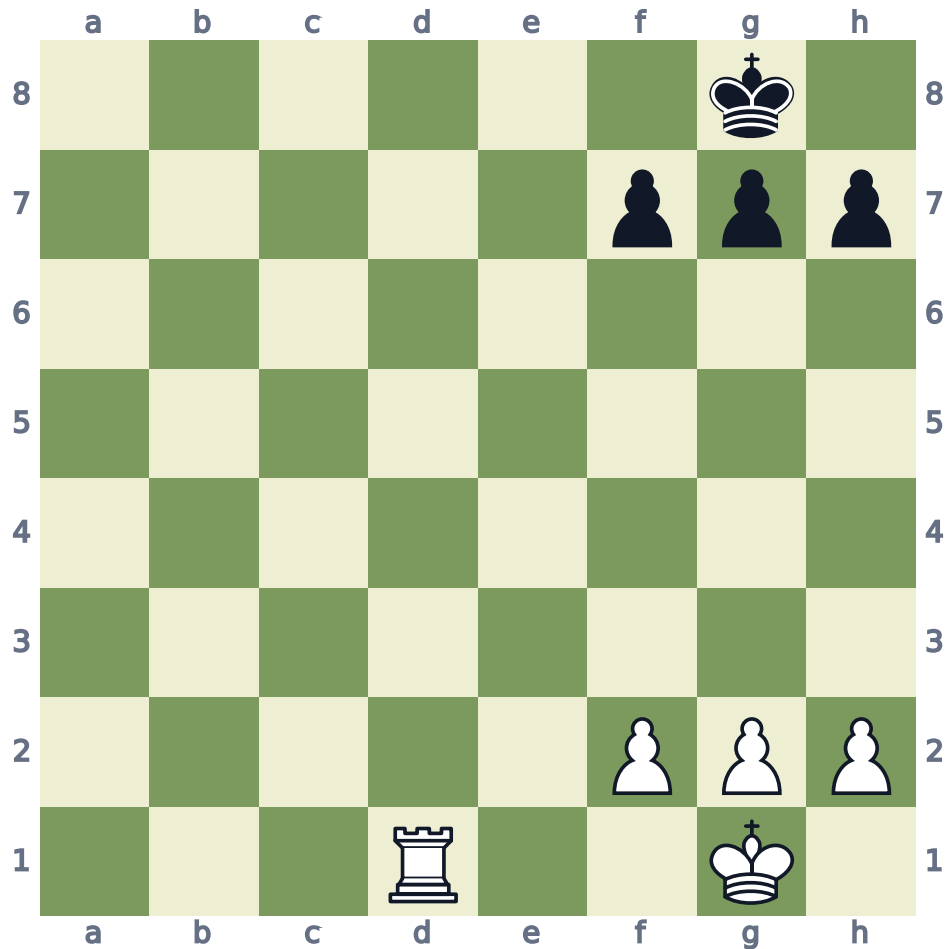
Stop only after forcing play settles, then evaluate material, kings, activity, and structure.

TRANSFER

State the visible cue that should make you remember this position during a game.

PROBLEM 01 OF 24

Calculation test 1: Back-rank mate



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

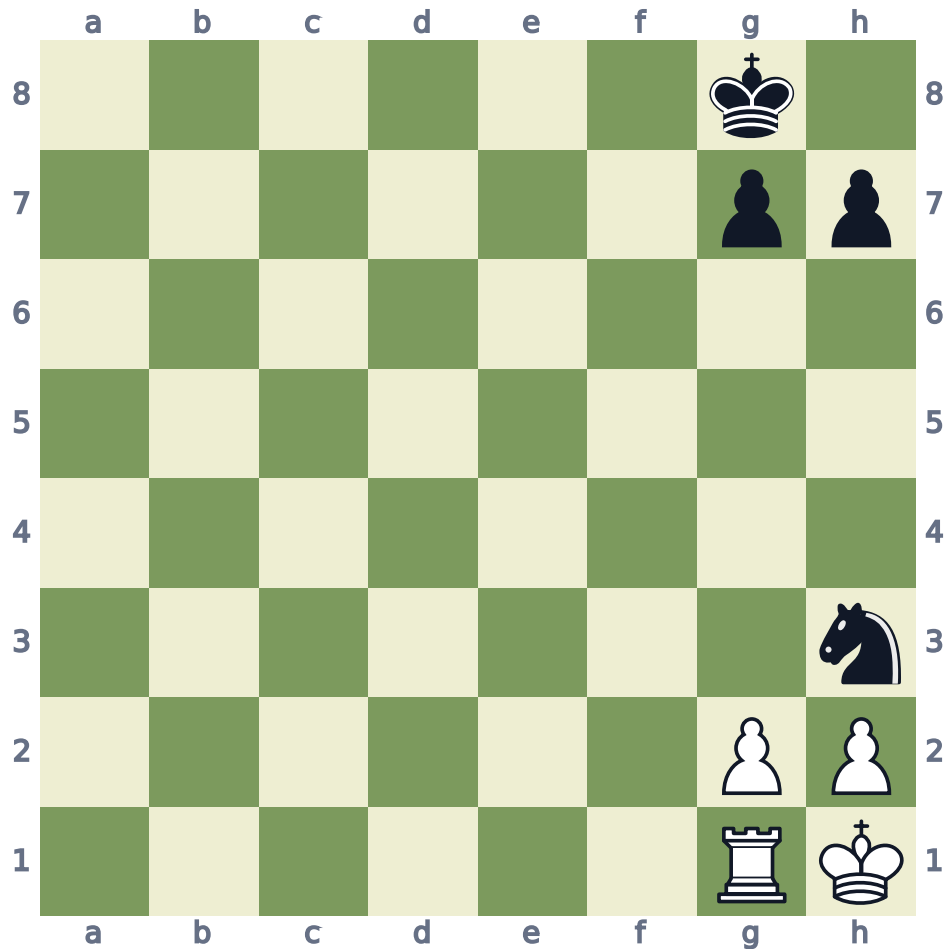
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 02 OF 24

Calculation test 2: Smothered mate - reversed colors



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

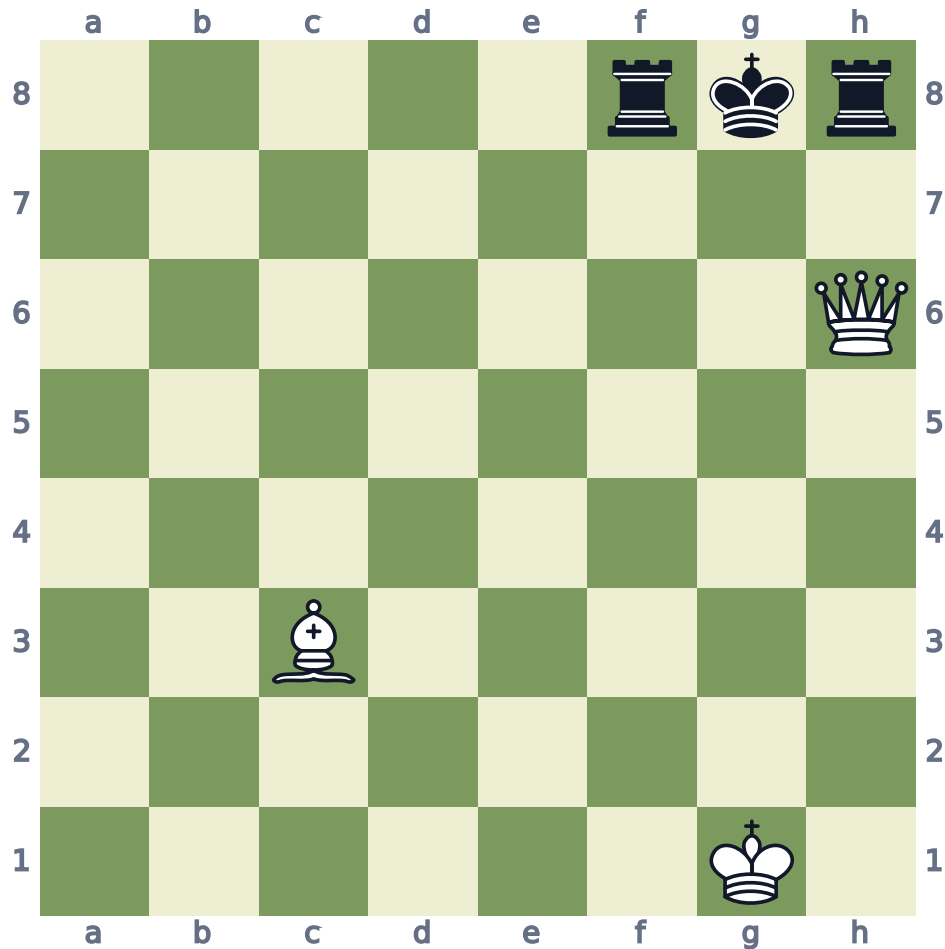
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 03 OF 24

Calculation test 3: Epaulette mate



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

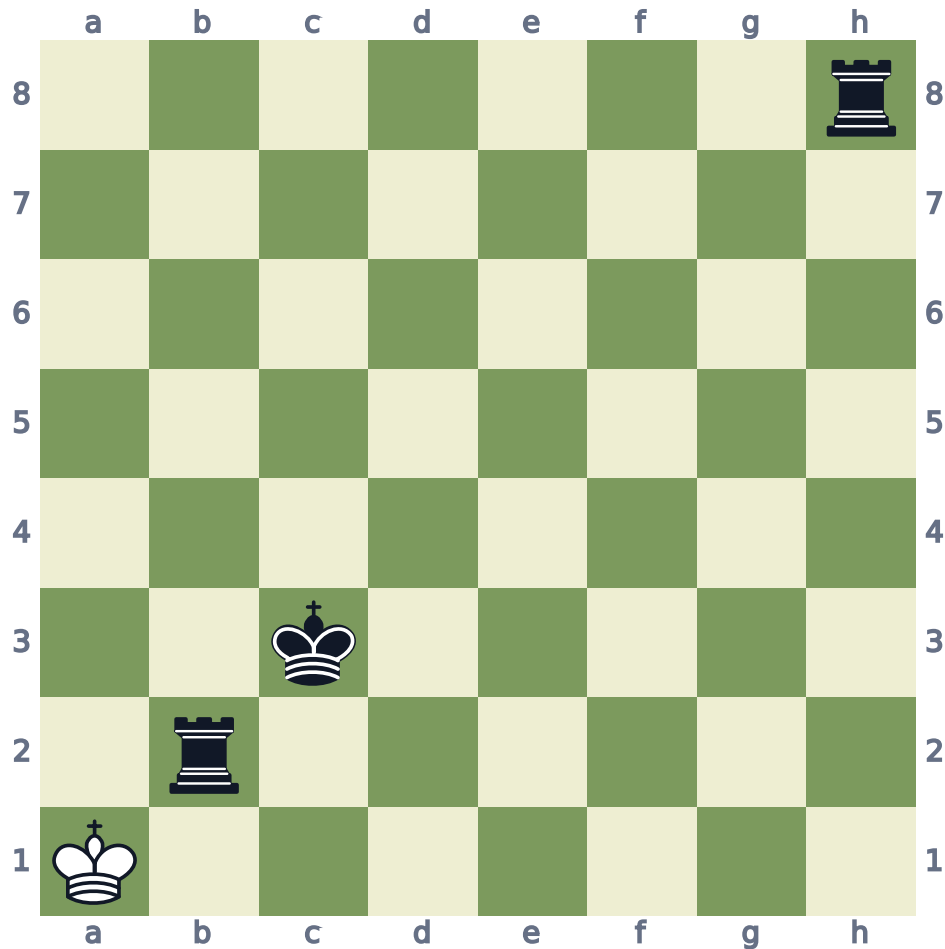
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 04 OF 24

Calculation test 4: Ladder mate - reversed colors



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

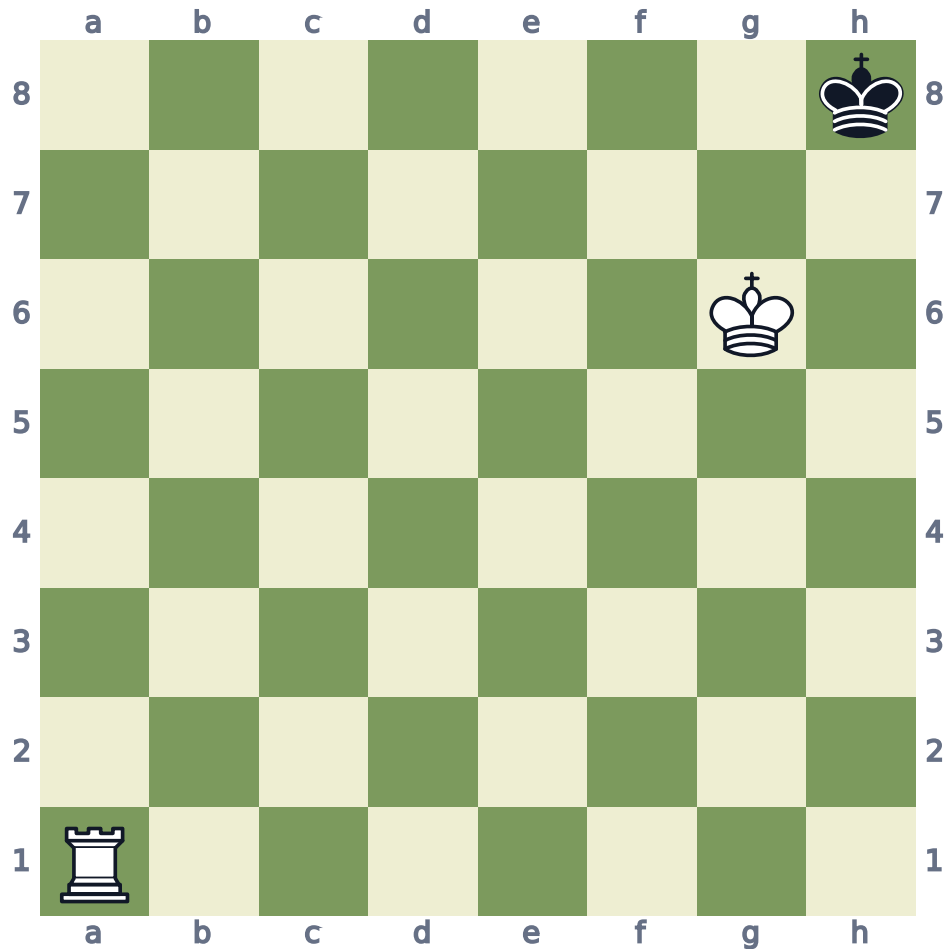
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 05 OF 24

Calculation test 5: Rook box mate



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

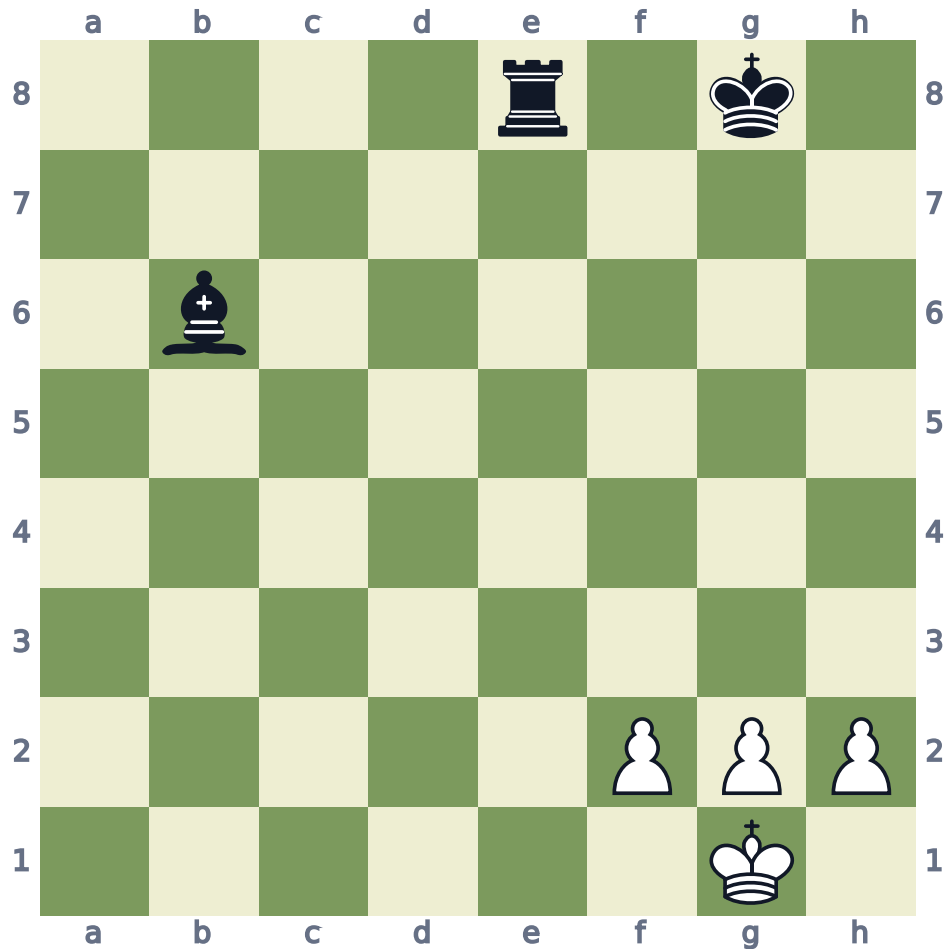
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 06 OF 24

Calculation test 6: Rook-and-bishop back rank - reversed colors



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

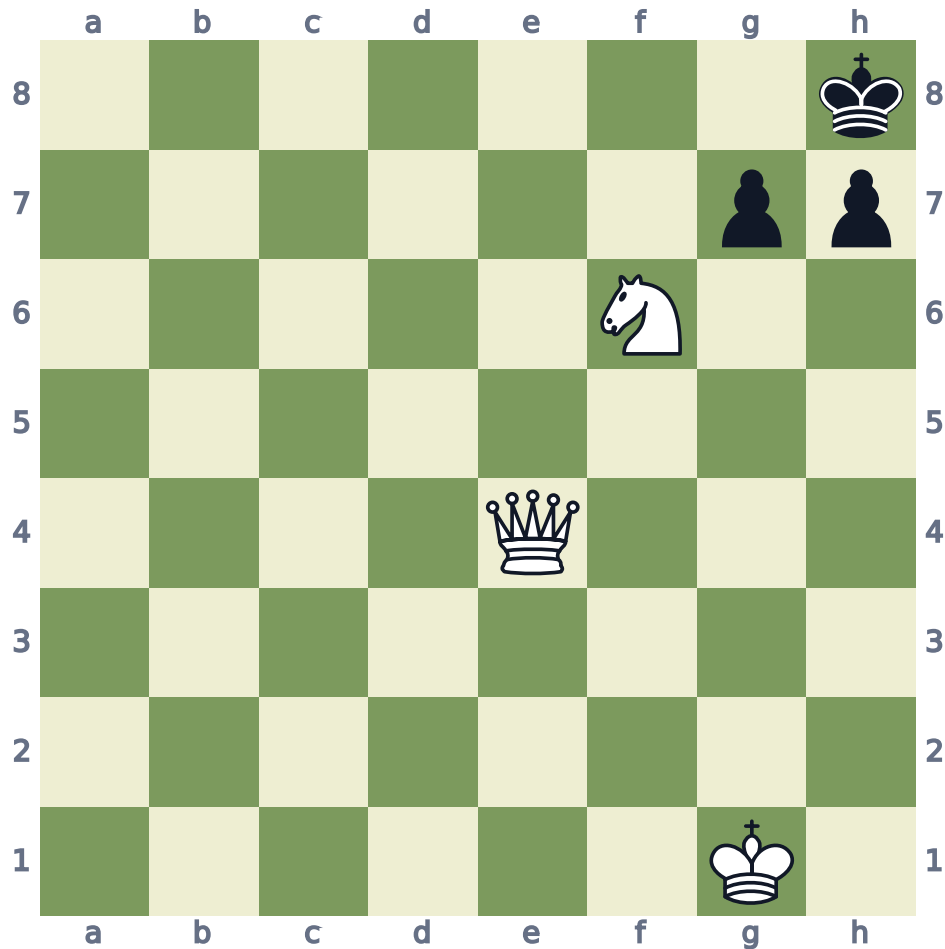
Two-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 07 OF 24

Calculation test 7: Queen-and-knight mate



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

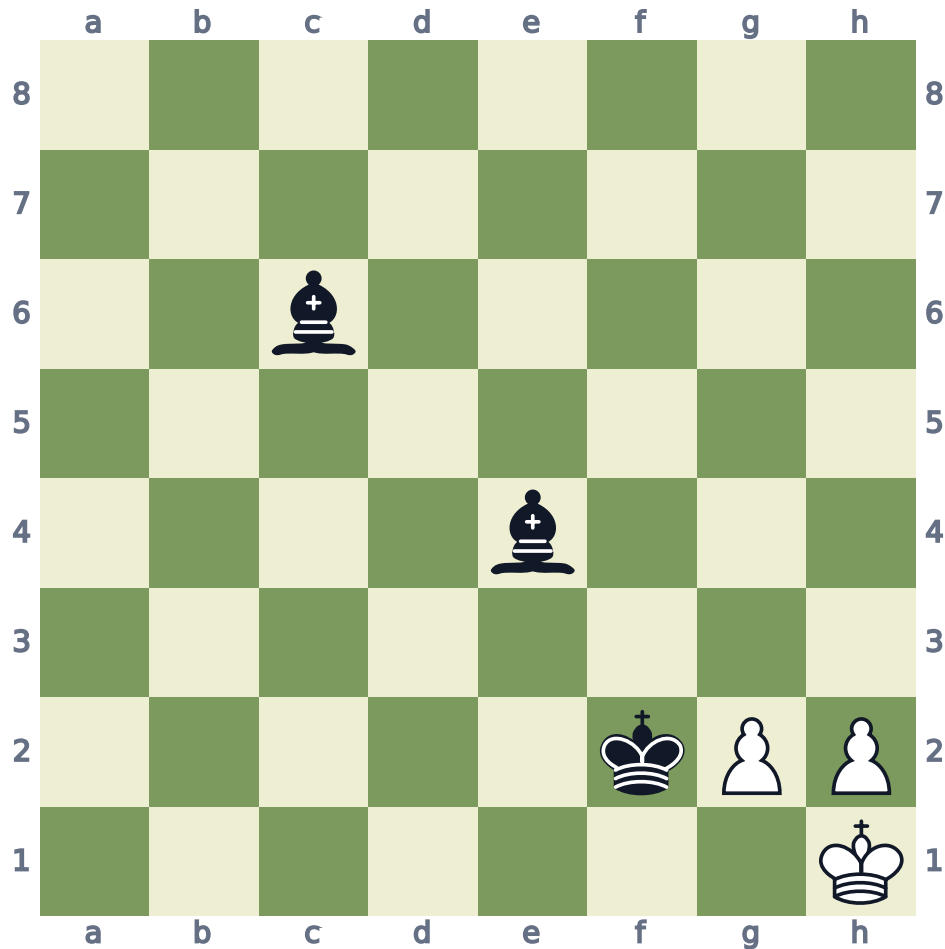
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 08 OF 24

Calculation test 8: Double-bishop mate - reversed colors



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

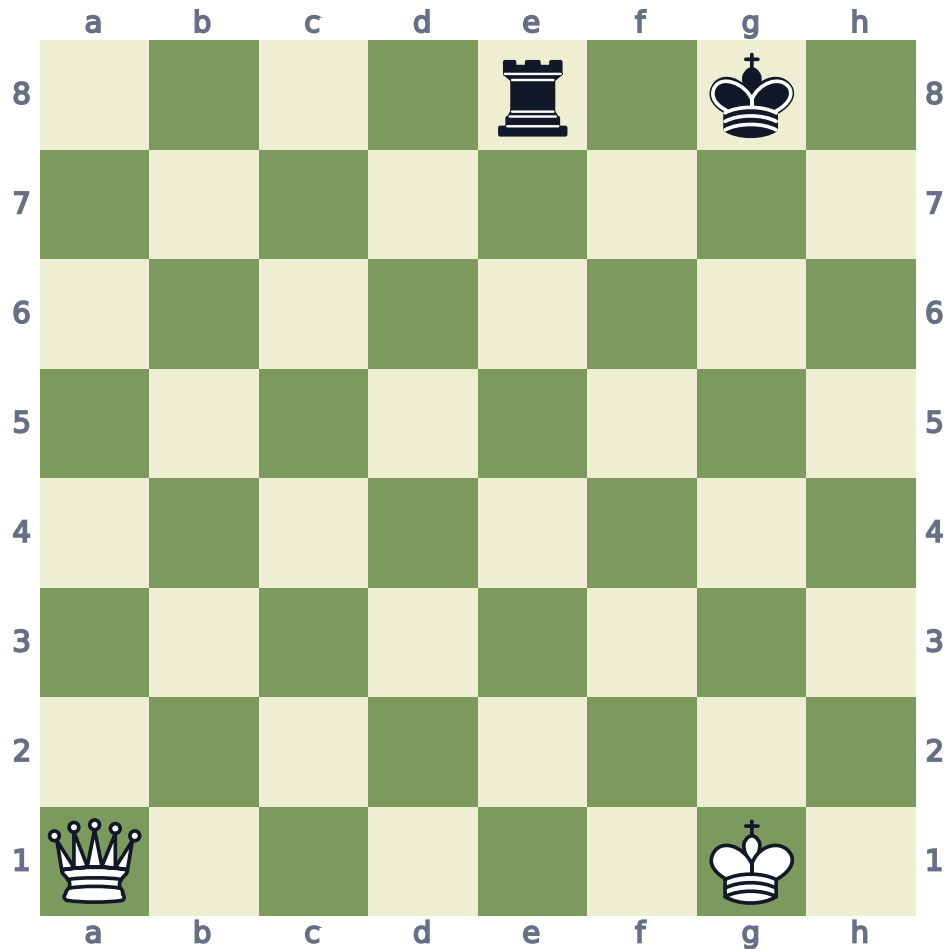
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 09 OF 24

Calculation test 9: Rook skewer - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

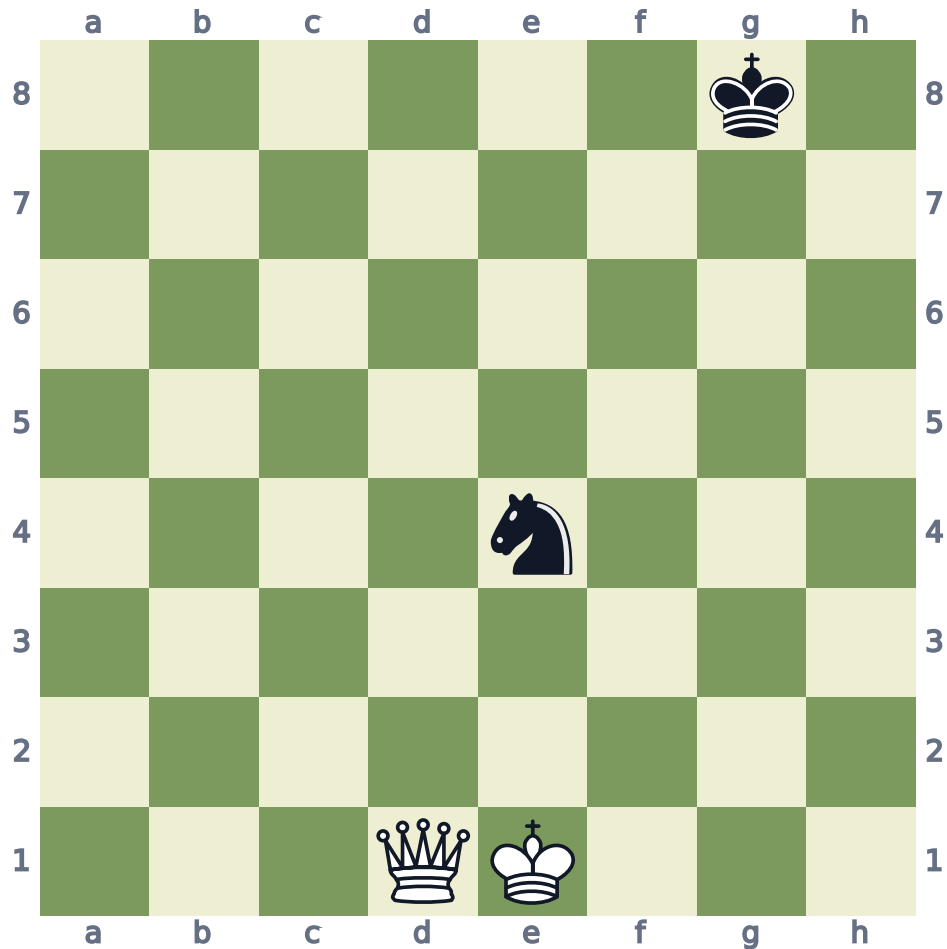
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 10 OF 24

Calculation test 10: Knight double attack - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

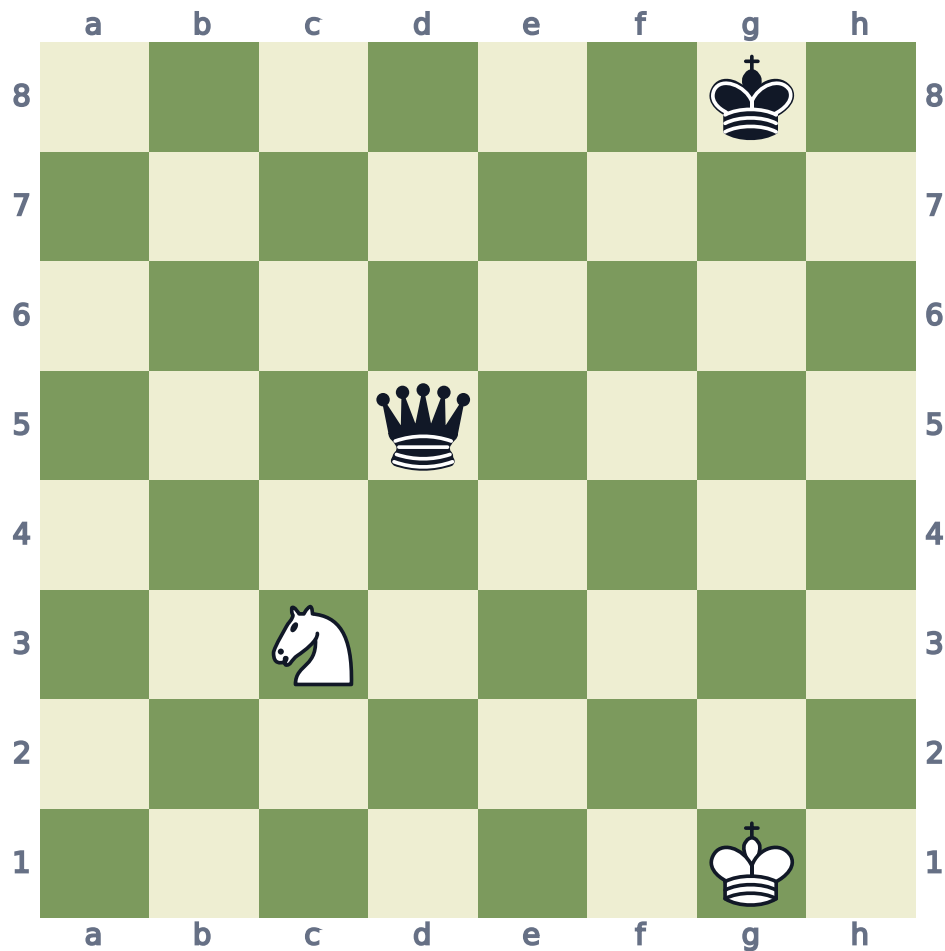
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 11 OF 24

Calculation test 11: Loose queen



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

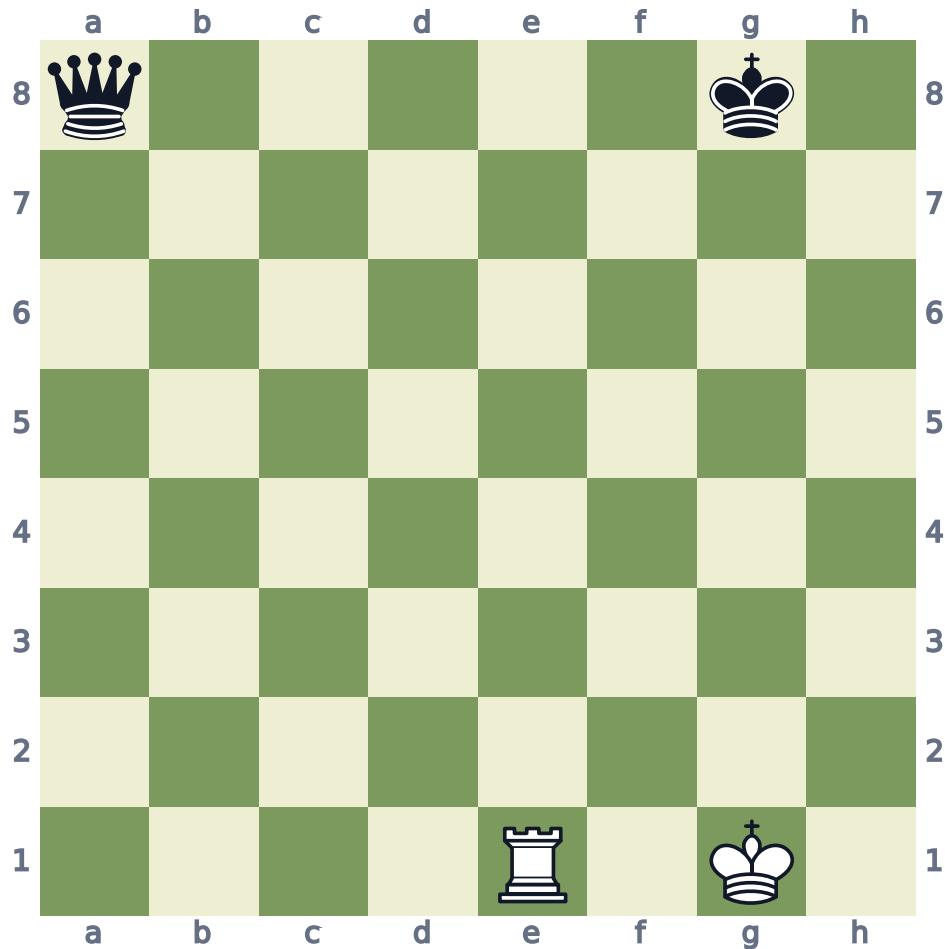
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 12 OF 24

Calculation test 12: Rook skewer



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

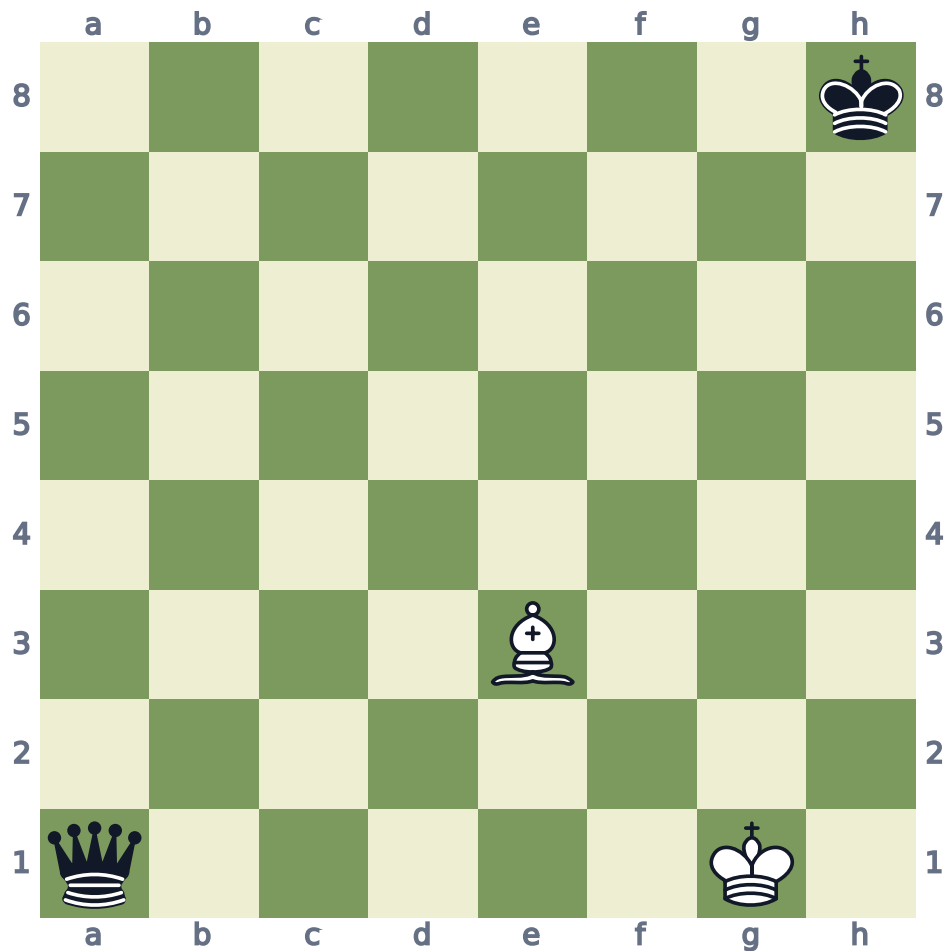
Five-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 13 OF 24

Calculation test 13: Bishop skewer



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

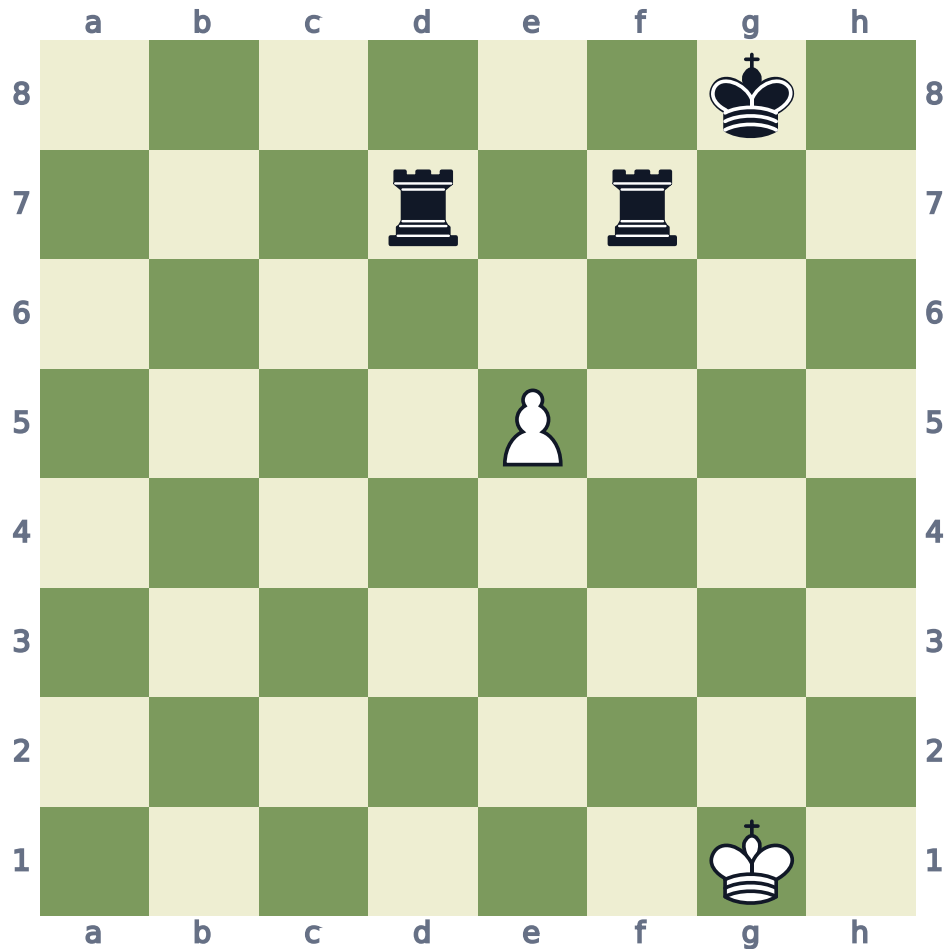
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 14 OF 24

Calculation test 14: Pawn fork



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

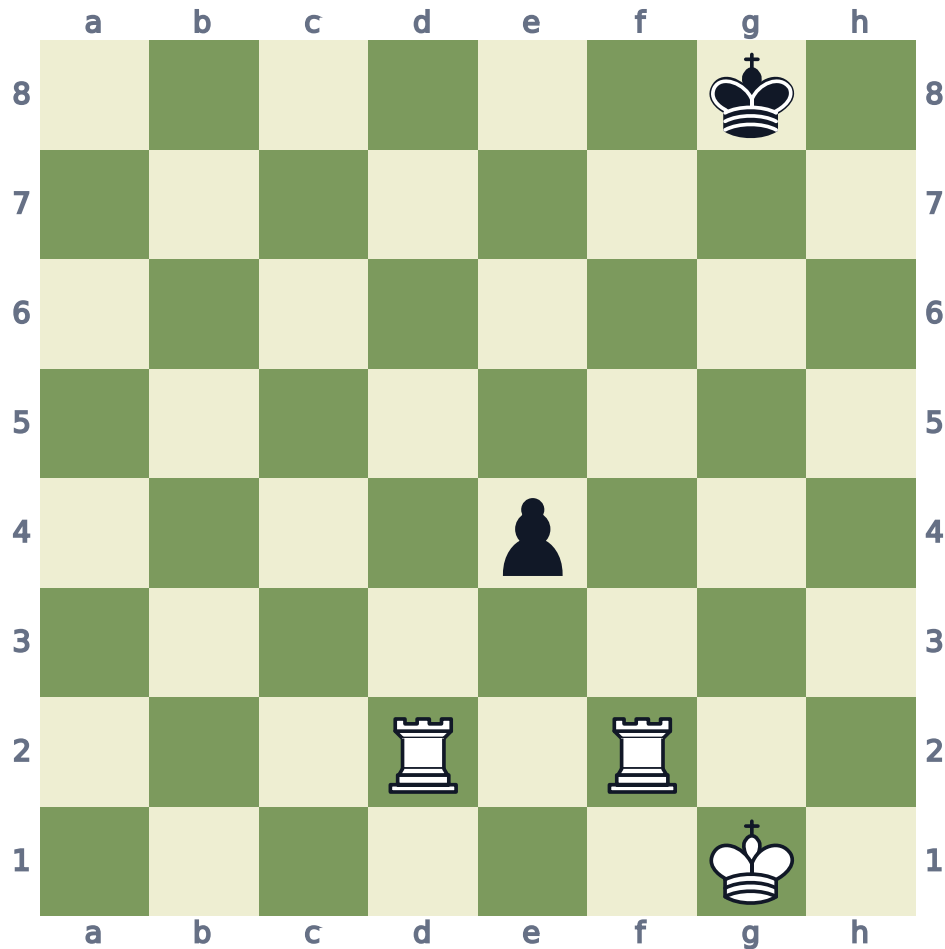
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 15 OF 24

Calculation test 15: Pawn fork - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

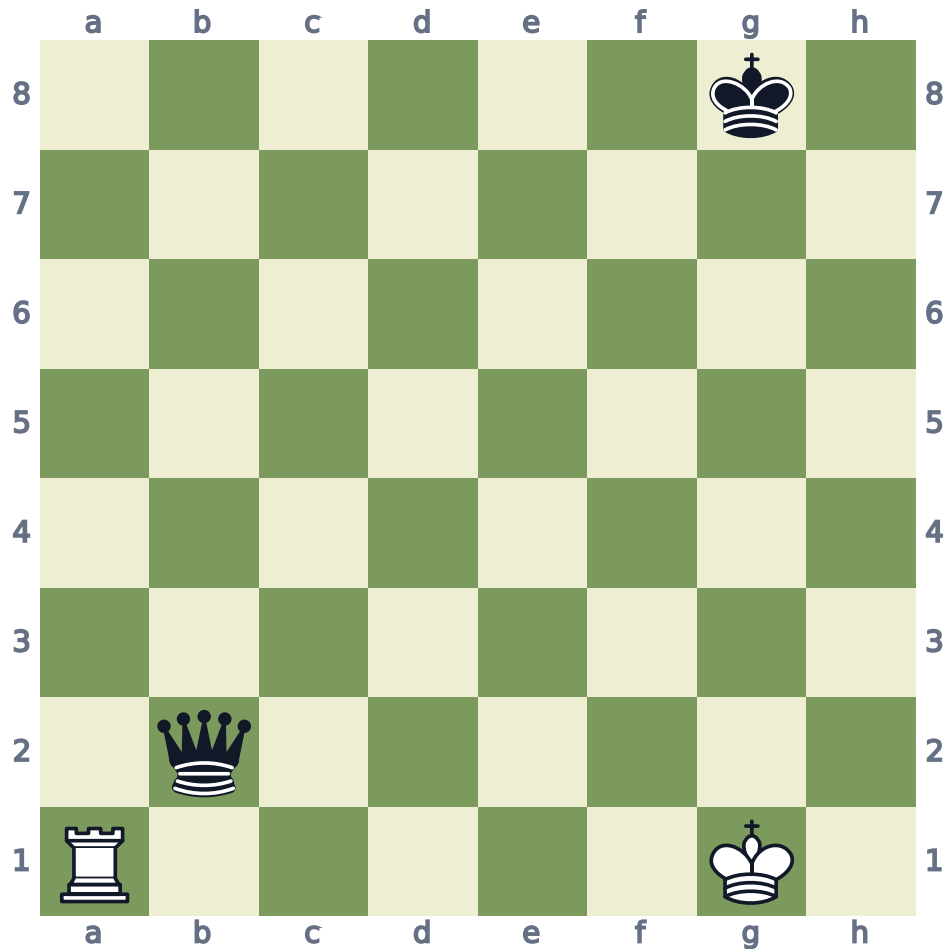
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 16 OF 24

Calculation test 16: Capture with check - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

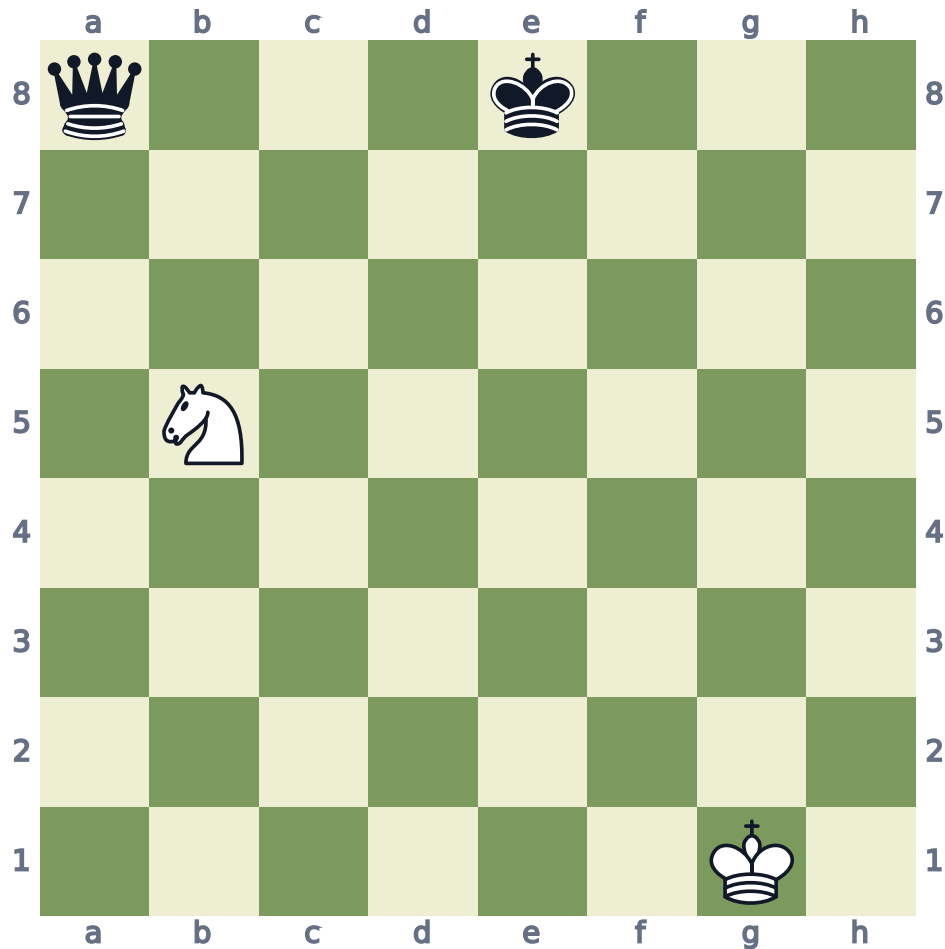
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 17 OF 24

Calculation test 17: Royal fork



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

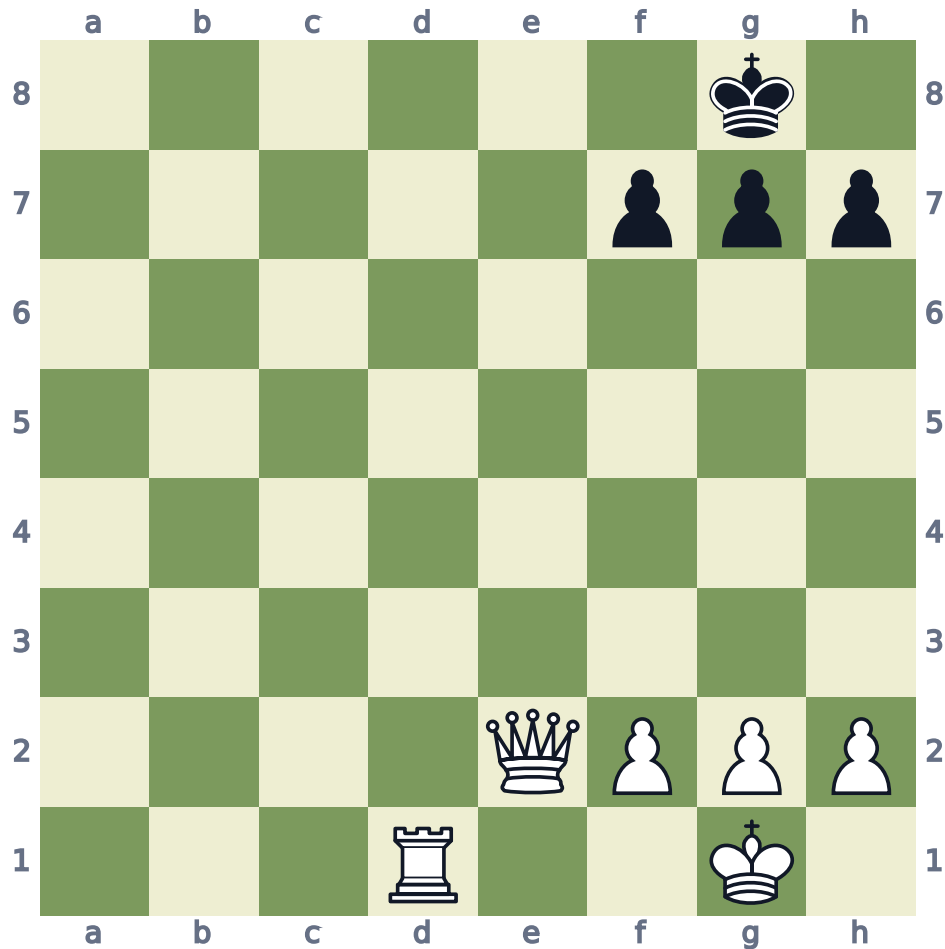
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 18 OF 24

Calculation test 18: Back-rank geometry



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

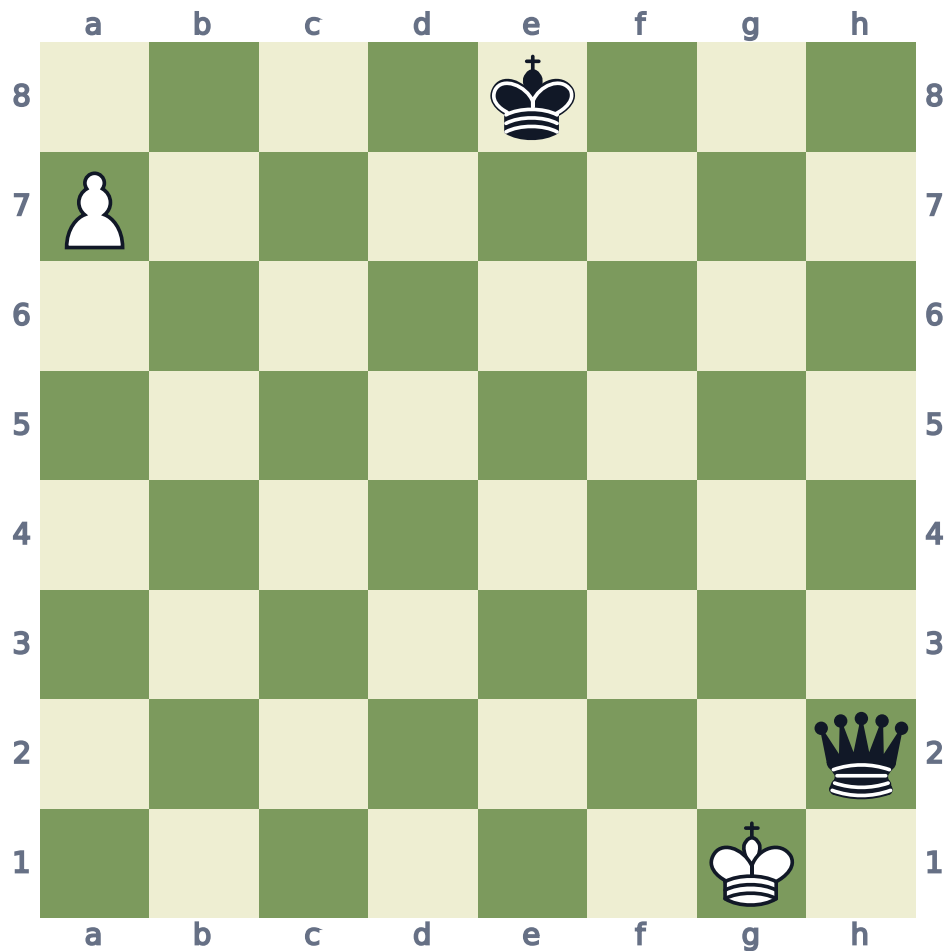
Ten-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 19 OF 24

Calculation test 19: Promotion tactic



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

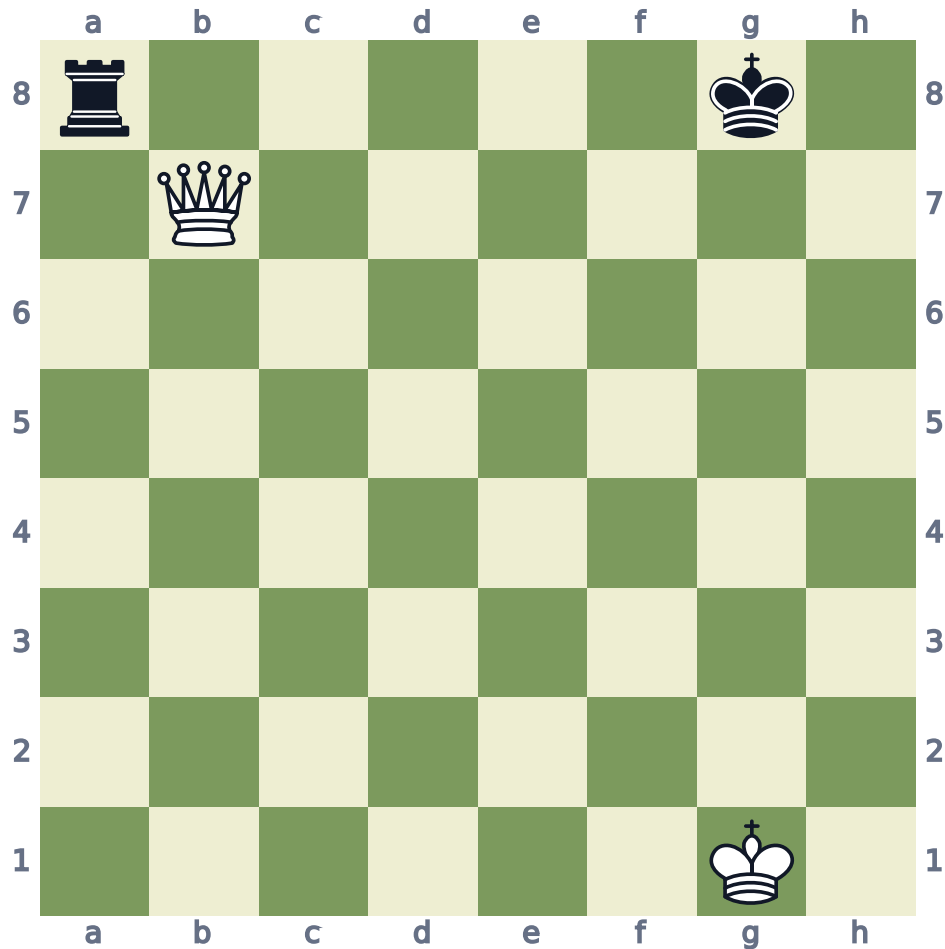
Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 20 OF 24

Calculation test 20: Capture with check



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

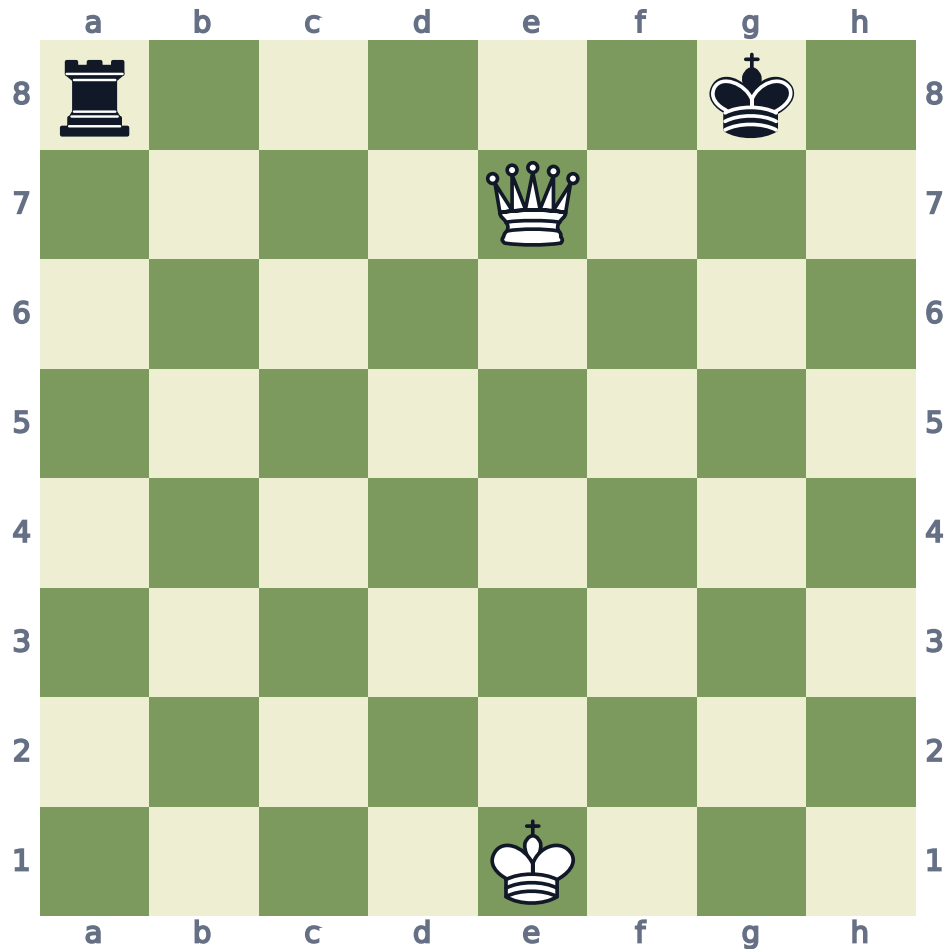
Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 21 OF 24

Calculation test 21: Rook double attack - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

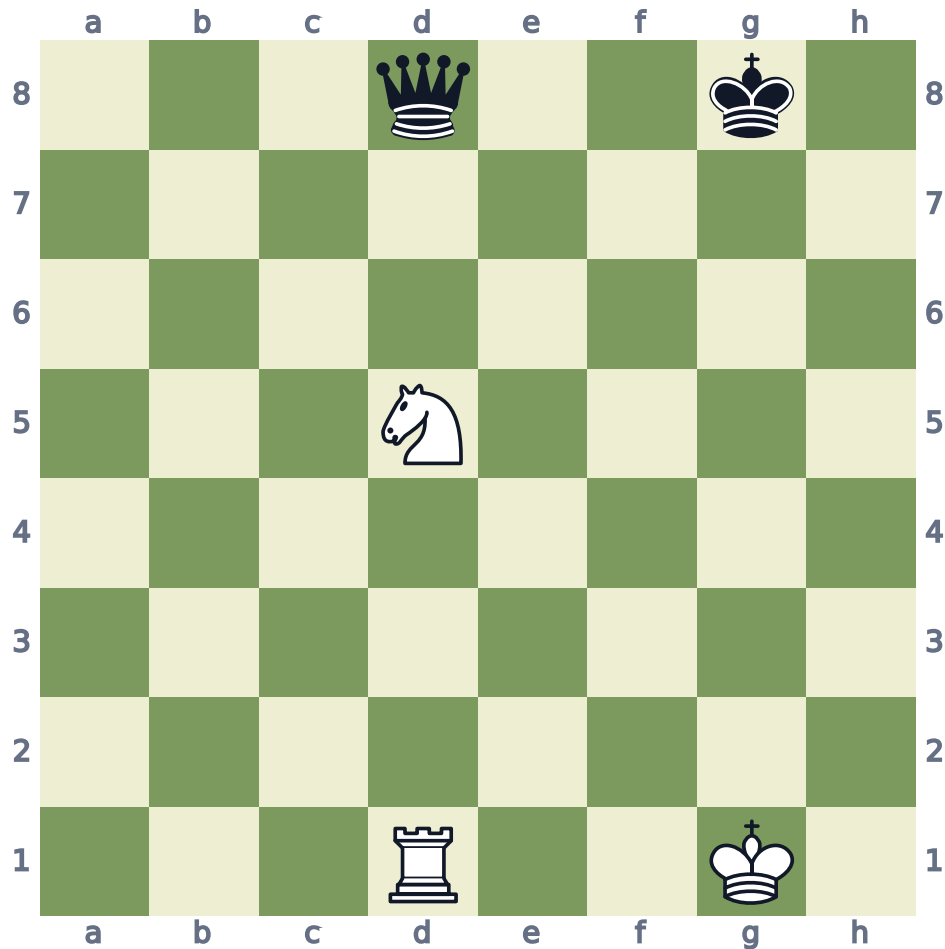
Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 22 OF 24

Calculation test 22: Discovered attack



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

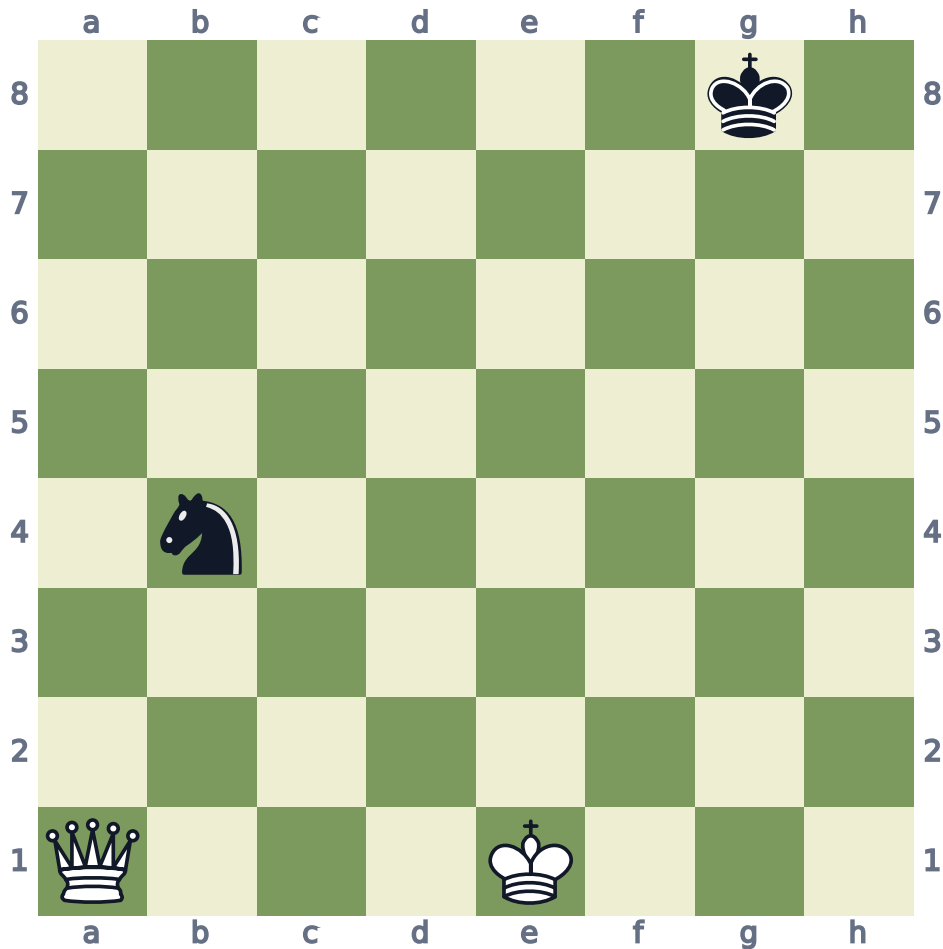
Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 23 OF 24

Calculation test 23: Royal fork - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

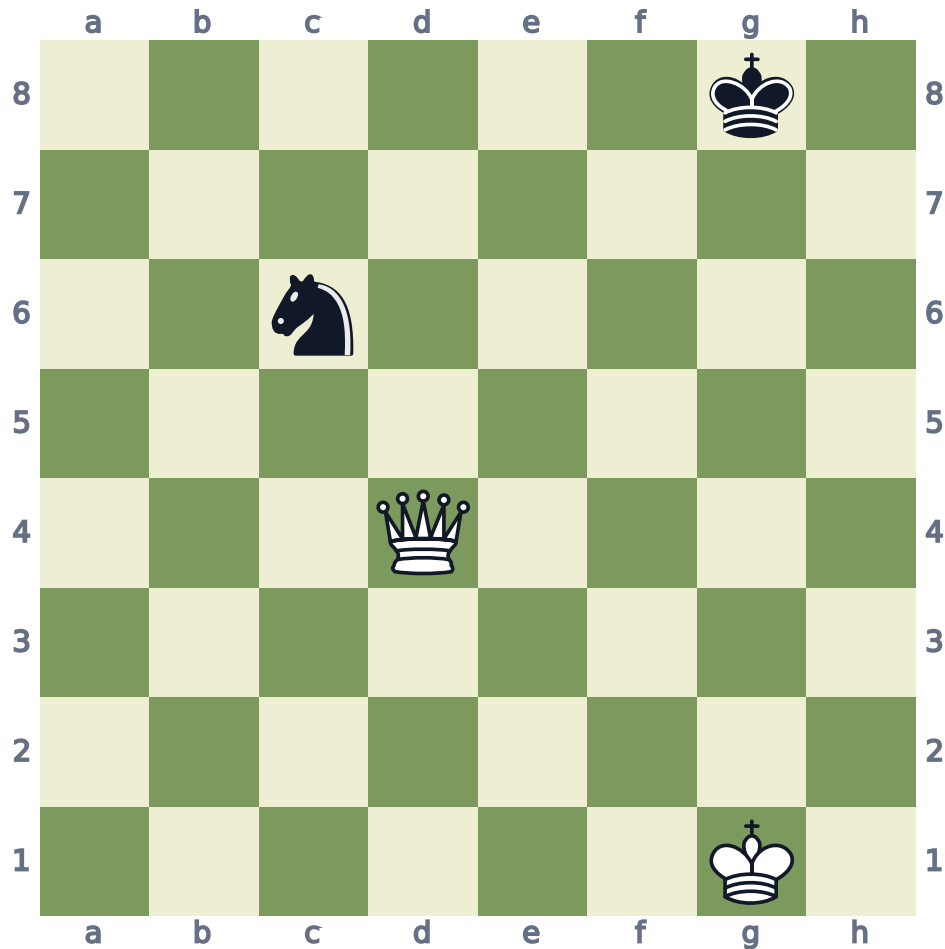
Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

PROBLEM 24 OF 24

Calculation test 24: Loose queen - reverse view



YOUR TASK

Write exactly three candidates. Calculate each until checks, major captures, and direct threats settle. Give a one-sentence endpoint evaluation.

DIFFICULTY

Fifteen-minute test

BEFORE TURNING TO THE SOLUTION SECTION

Record candidates, the best defensive reply, and your final evaluation. A correct first move without a verified line earns only partial credit.

ANSWER SECTION

Solutions begin on the next page

PROTECT THE TRAINING EFFECT

Do not browse the solution pages casually. Return to each problem, reconstruct the position, and compare the entire decision process. If your move matches but your line omits the best defense, score it as incomplete.

ERROR LABELS

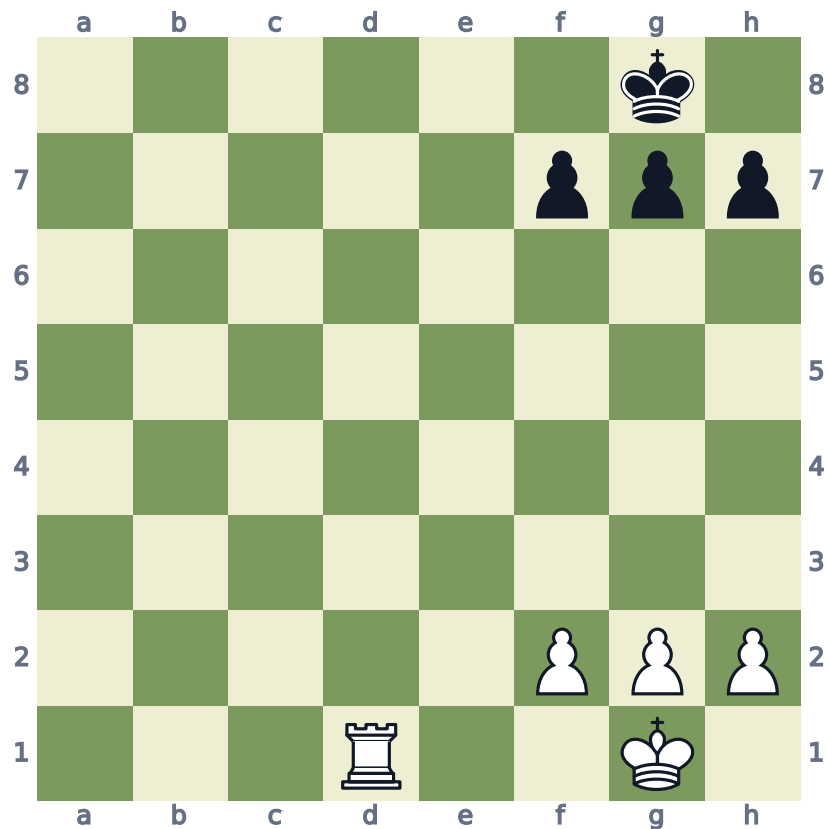
R = recognition error | C = candidate error | V = visualization error | D = missed defense | E = endpoint evaluation error | T = time-management error

REPAIR RULE

Every miss must produce one replacement question. Example: 'Before I calculate my attack, what countercheck does the opponent have?' Use that question in the next five slow games.

SOLUTION 01 OF 24

Calculation test 1: Back-rank mate



VERIFIED LINE

1. Rd8#

WHY IT WORKS

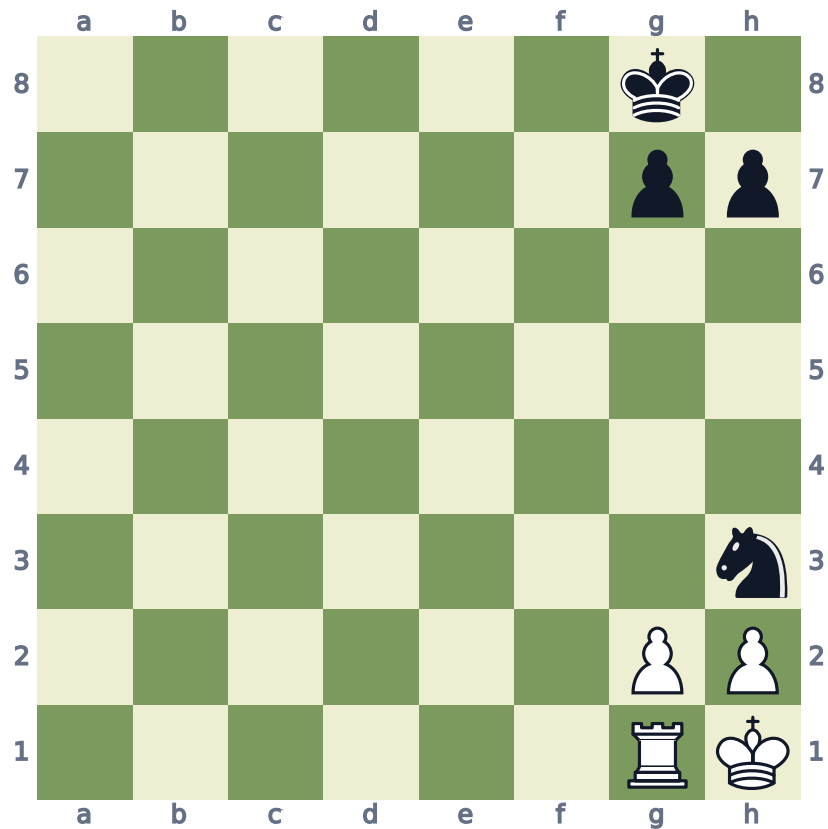
The best line is compared against these alternatives: 1. Kh1 f5; 1. g4 h6. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 02 OF 24

Calculation test 2: Smothered mate - reversed colors



VERIFIED LINE

1. Nf2#

WHY IT WORKS

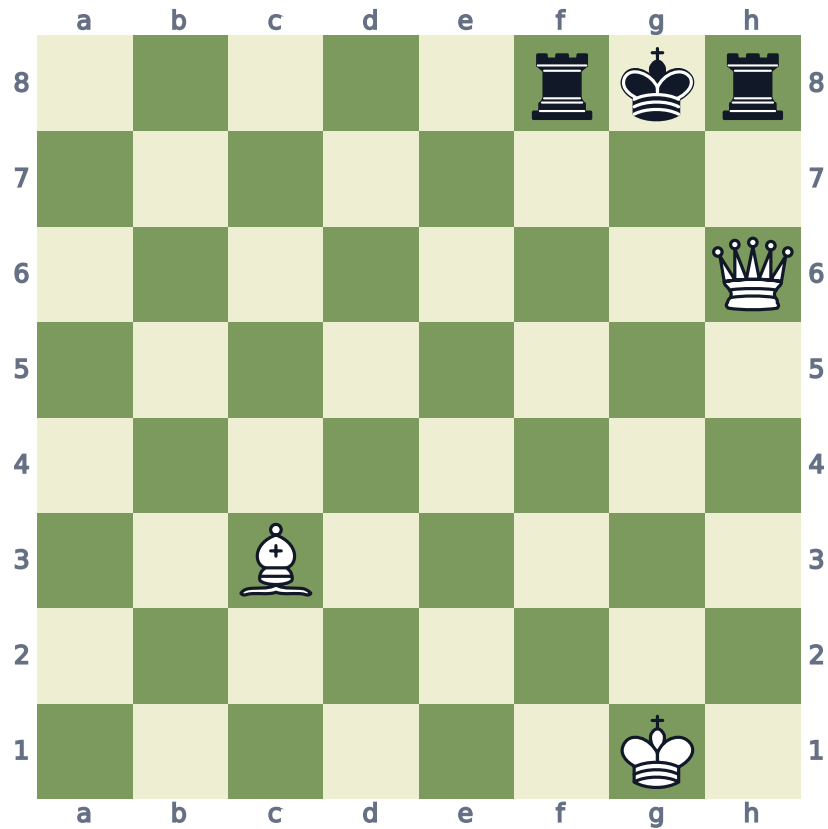
The best line is compared against these alternatives: 1. Nxg1 Kxg1 2. Kf7 g4 3. h6 Kf2 4. Ke6 Ke3 5. Kd5 Kf4; 1. Nf4 Rb1 2. Nd5 Rb8+ 3. Kf7 Rb7+ 4. Kf6. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 03 OF 24

Calculation test 3: Epaulette mate



VERIFIED LINE

1. Qg7#

WHY IT WORKS

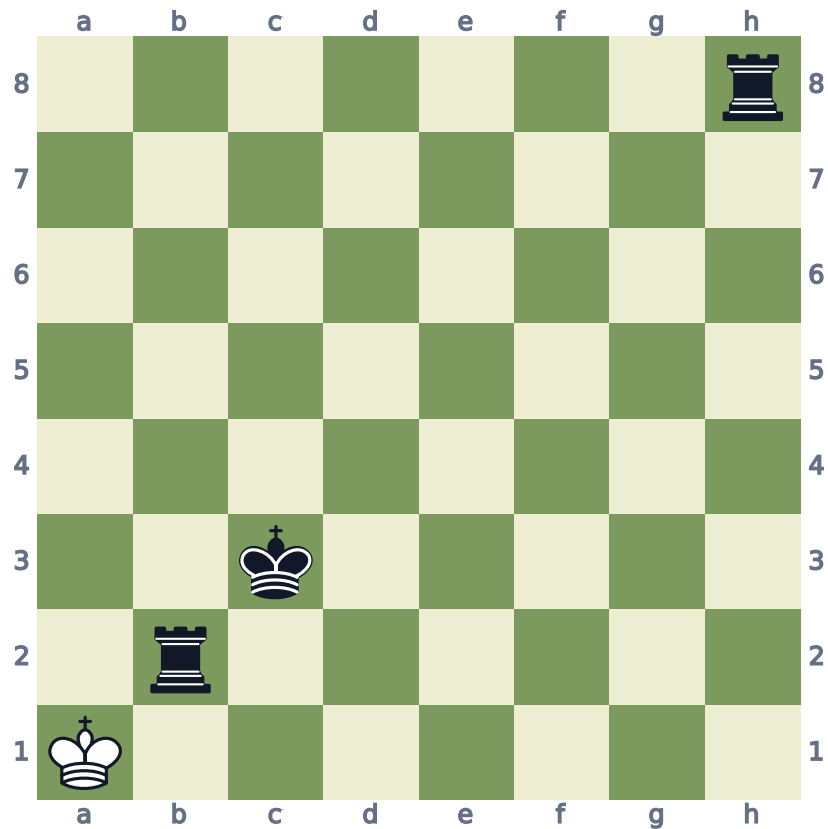
The best line is compared against these alternatives: 1. Qg6#; 1. Qxh8+ Kf7 2. Qf6+ Ke8 3. Qe6+ Kd8 4. Ba5#. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 04 OF 24

Calculation test 4: Ladder mate - reversed colors



VERIFIED LINE

1. Rh1#

WHY IT WORKS

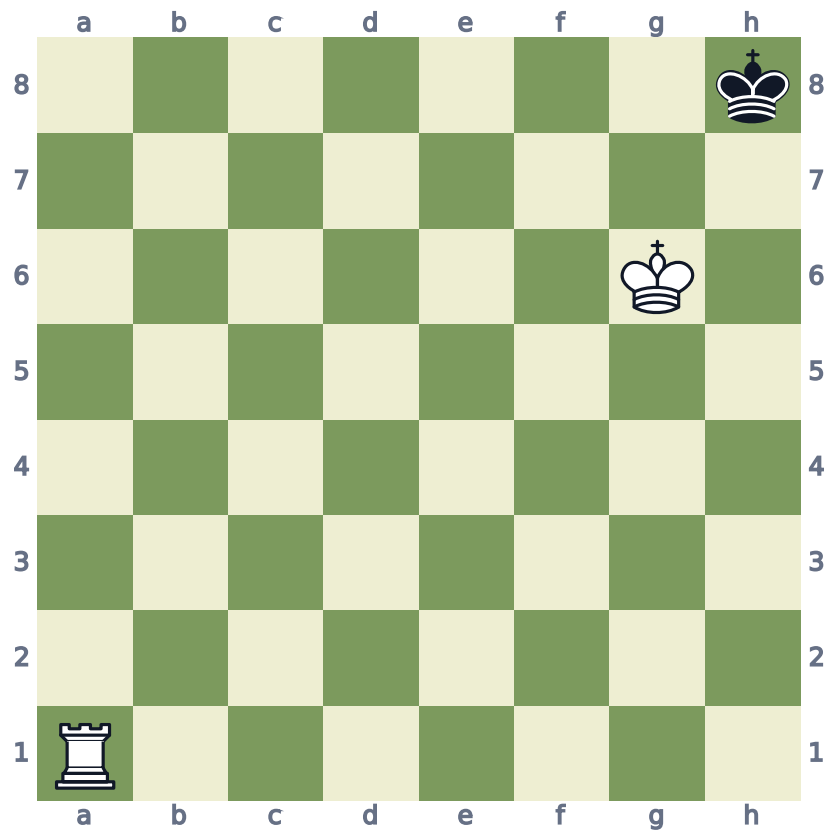
The best line is compared against these alternatives: 1. Ra8#; 1. Rf2 Kb1 2. Rh1#. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 05 OF 24

Calculation test 5: Rook box mate



VERIFIED LINE

1. Ra8#

WHY IT WORKS

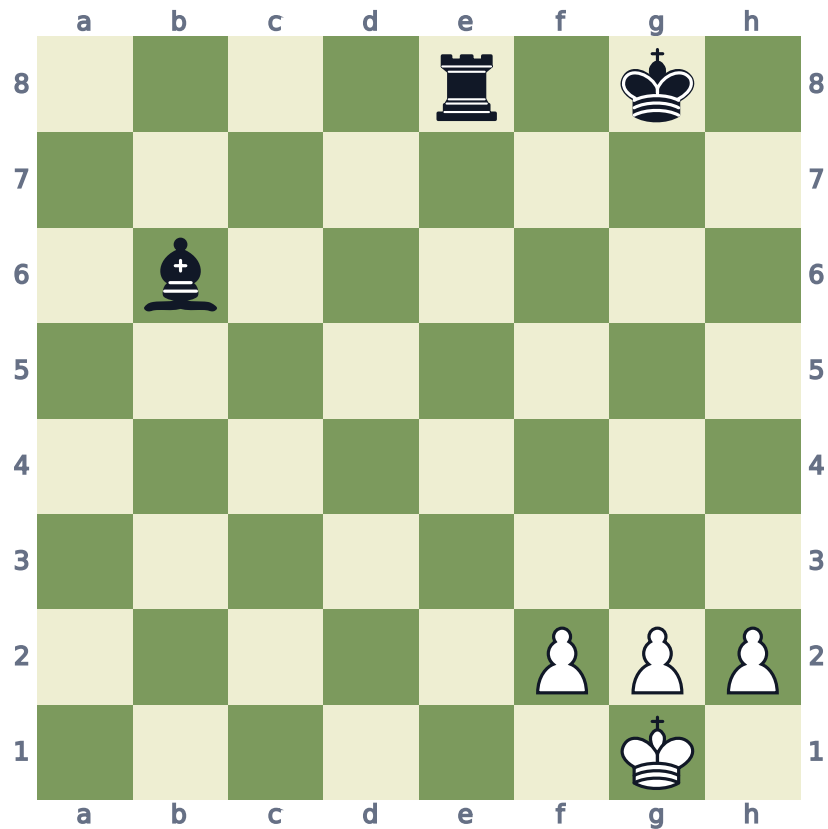
The best line is compared against these alternatives: 1. Rb1 Kg8 2. Rb8#; 1. Rc1 Kg8 2. Rc8#. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 06 OF 24

Calculation test 6: Rook-and-bishop back rank - reversed colors



VERIFIED LINE

1. Re1#

WHY IT WORKS

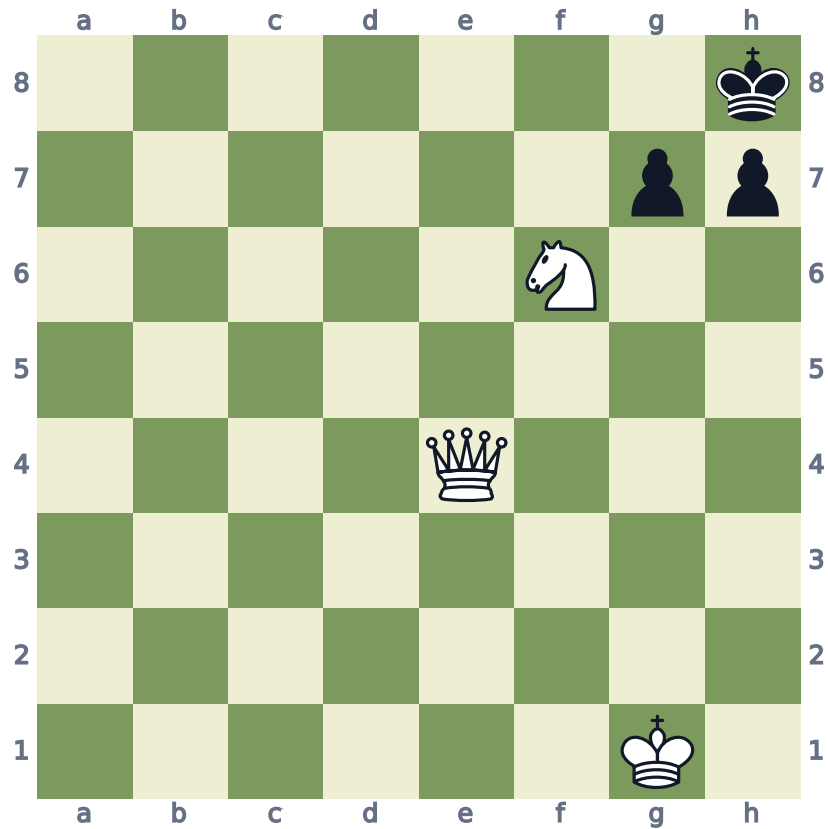
The best line is compared against these alternatives: 1. Re2 g4 2. Rxf2 Kh1 3. Rf7 Kg2 4. Ra7 h3 5. Bc7 Kg1; 1. Rf8 Kh1 2. Bxf2 g3 3. Rf7 g4 4. Be1 Kg2 5. Ra7 g5. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 07 OF 24

Calculation test 7: Queen-and-knight mate



VERIFIED LINE

1. Qxh7#

WHY IT WORKS

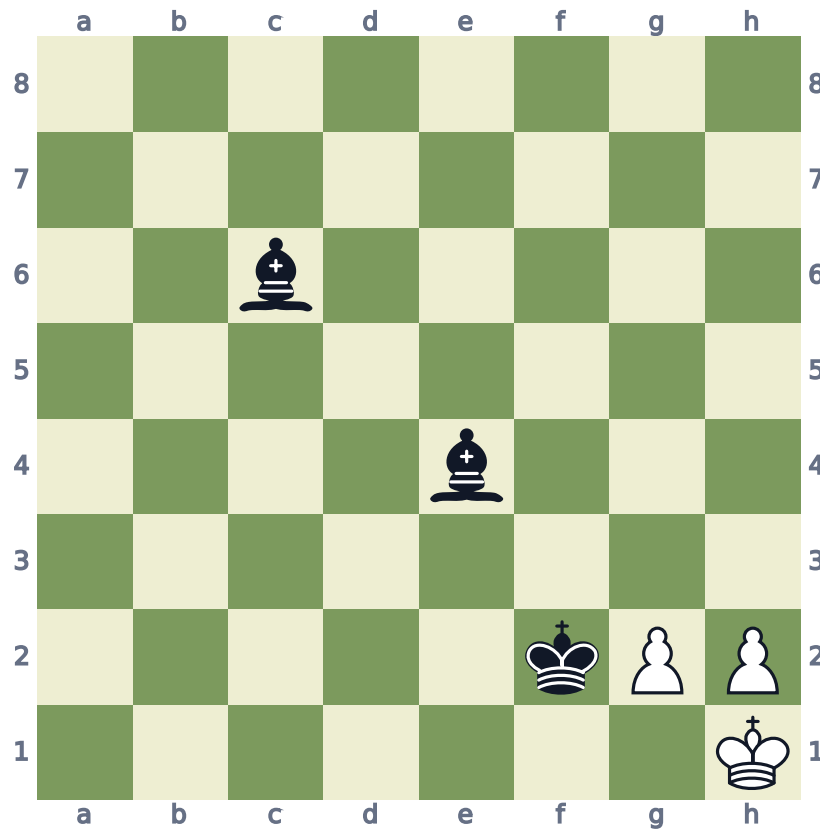
The best line is compared against these alternatives: 1. Qa8#; 1. Qe8#. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 08 OF 24

Calculation test 8: Double-bishop mate - reversed colors



VERIFIED LINE

1. Bxg2#

WHY IT WORKS

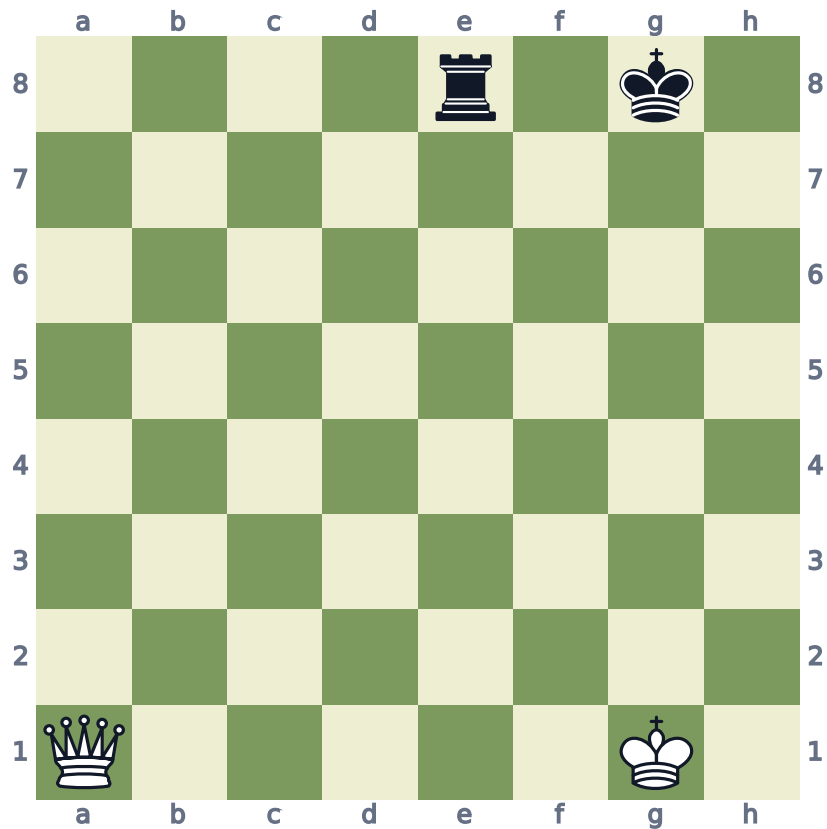
The best line is compared against these alternatives: 1. Bd5 h4 2. Bxg2+ Kh2 3. Bf1 h5 4. Bd5 h6 5. Be4 h7; 1. Bb5 h4 2. Bxg2+ Kh2 3. Bbf1 h5 4. Bb7 h6 5. Be4 h7. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 09 OF 24

Calculation test 9: Rook skewer - reverse view



VERIFIED LINE

1. Rf8 Kh1 2. Rd8 Kg2 3. Kf8 Qb1 4. Ke7 Kf3 5. Rd6 Kf4

WHY IT WORKS

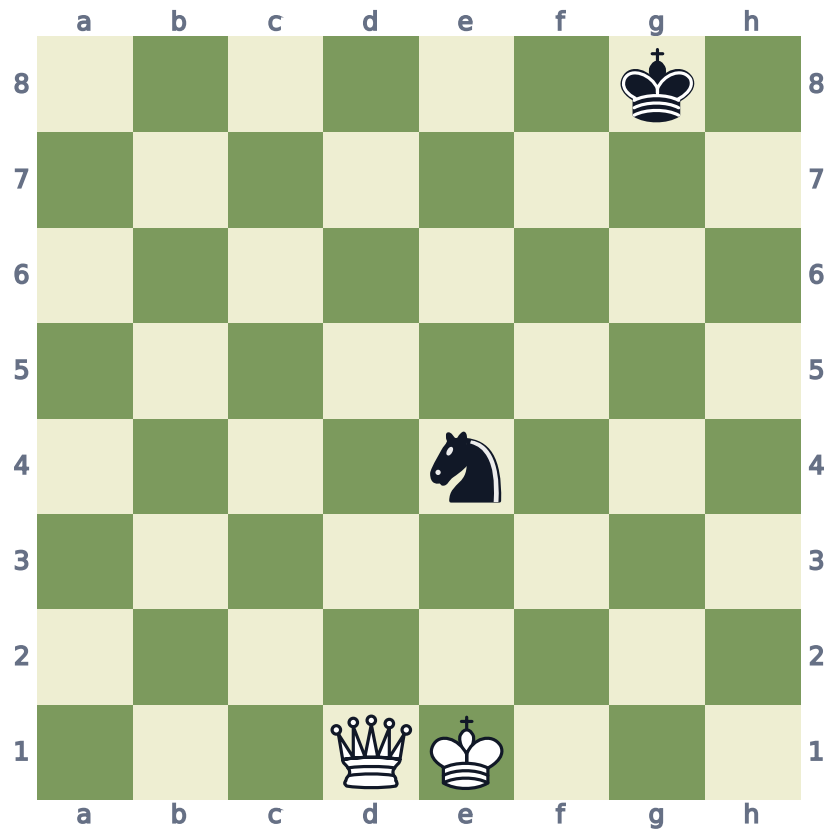
The best line is compared against these alternatives: 1. Kf7 Qa2+ 2. Re6 Kf2 3. Ke7 Kf3 4. Re1 Qd2 5. Rf1+ Ke4; 1. Rc8 Qa2+ 2. Kf8 Kg2 3. Ke7 Kf3 4. Rc7 Ke4 5. Kd8 Qg8+. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -2.36 from the side-to-move perspective.

SOLUTION 10 OF 24

Calculation test 10: Knight double attack - reverse view



VERIFIED LINE

1. Kf7 Qd5+ 2. Ke7 Qe5+ 3. Kd7 Qxe4 4. Kd6 Qd4+ 5. Ke6 Qc5

WHY IT WORKS

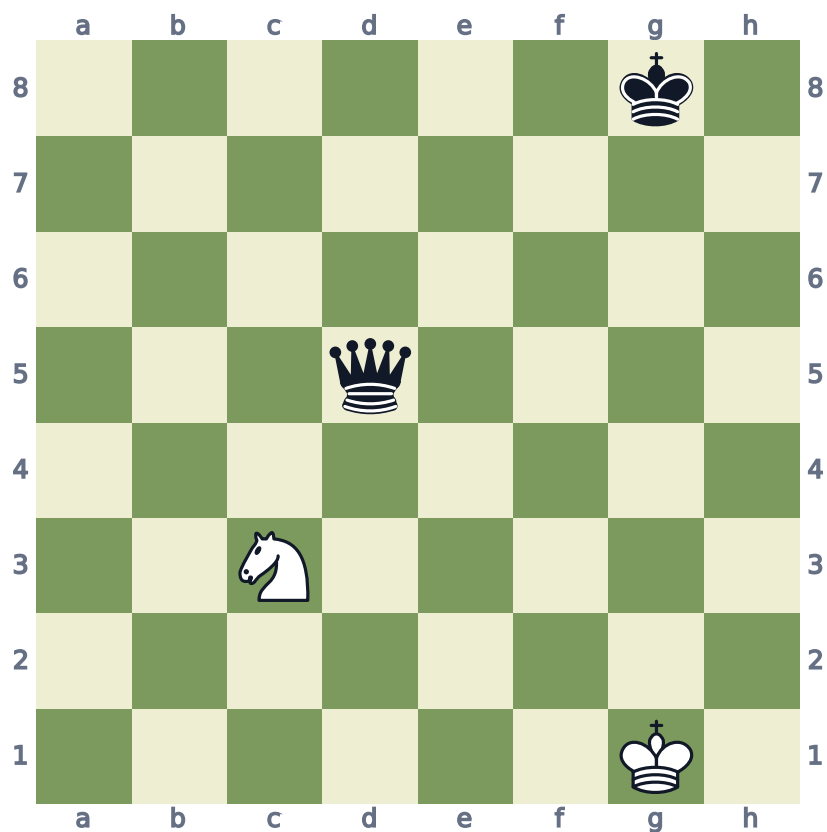
The best line is compared against these alternatives: 1. Nf6 Kf1 2. Kf7 Kg2 3. Ke6 Qe1+ 4. Kf5 Qf2+ 5. Ke4 Qh4+; 1. Kf8 Qd8+ 2. Kf7 Qd5+ 3. Ke7 Qe5+ 4. Kd7 Qxe4 5. Kd6 Kf1. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -4.92 from the side-to-move perspective.

SOLUTION 11 OF 24

Calculation test 11: Loose queen



VERIFIED LINE

1. Nxd5 Kf8 2. Ne3 Ke8 3. Ng4 Kd7 4. Kf1 Kd6

WHY IT WORKS

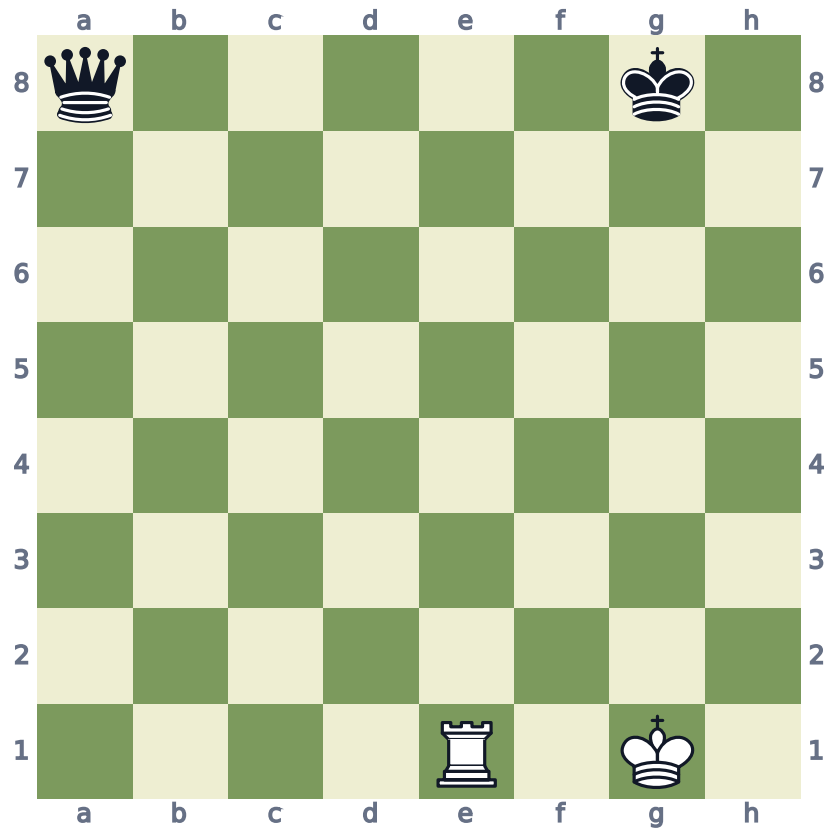
The best line is compared against these alternatives: 1. Kf2 Qc5+ 2. Kf3 Qxc3+ 3. Kf4 Kf7 4. Ke4 Qb2 5. Kf5; 1. Nb1 Qa2 2. Kf1 Qxb1+ 3. Ke2 Qb7 4. Ke3 Kg7 5. Ke2. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: +0.00 from the side-to-move perspective.

SOLUTION 12 OF 24

Calculation test 12: Rook skewer



VERIFIED LINE

1. Kf2 Kg7 2. Ke3 Kf6 3. Rc1 Qa7+ 4. Kd2 Ke5 5. Kd3 Qh7+

WHY IT WORKS

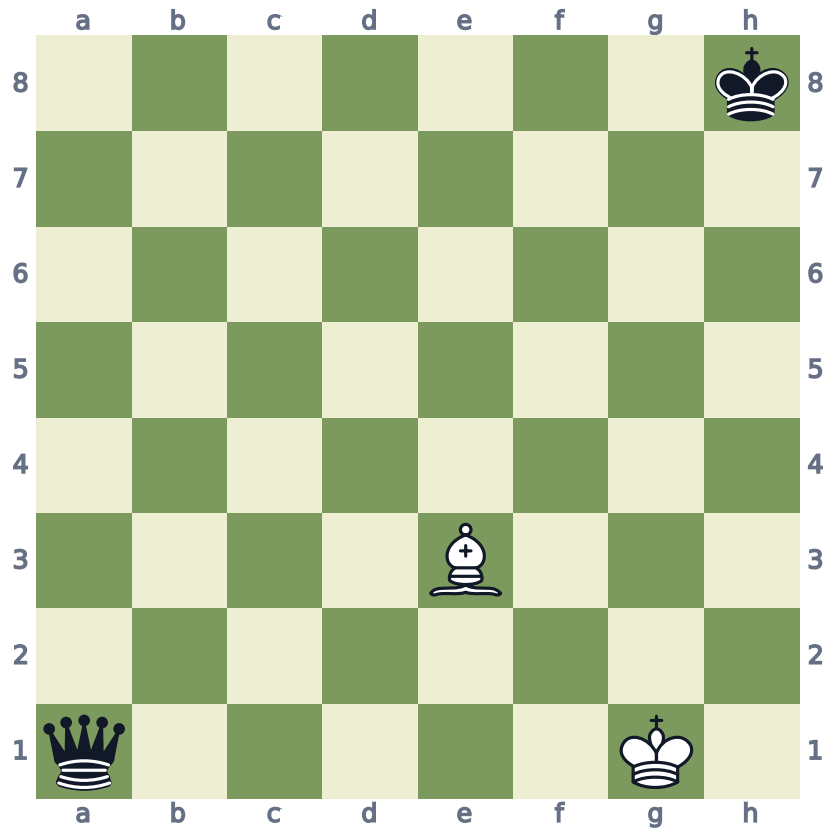
The best line is compared against these alternatives: 1. Kf1 Kg7 2. Re3 Qa7 3. Ke2 Kf6 4. Rd3 Ke5 5. Re3+ Kf4; 1. Re3 Kg7 2. Kf2 Kf6 3. Ke1 Kf5 4. Kd1 Kg4 5. Kd2 Kf4. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -2.24 from the side-to-move perspective.

SOLUTION 13 OF 24

Calculation test 13: Bishop skewer



VERIFIED LINE

1. Kg2 Kg7 2. Bg5 Qa6 3. Kf3 Qd3+ 4. Kf2 Qf5+ 5. Bf4 Kf6

WHY IT WORKS

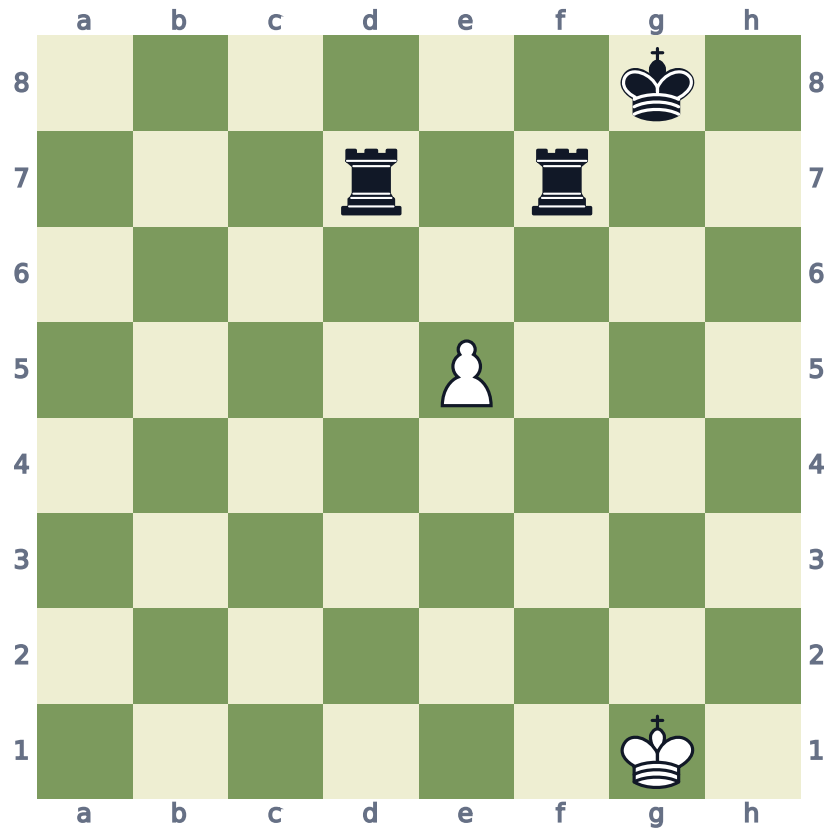
The best line is compared against these alternatives: 1. Kf2 Kg7 2. Bd2 Qa2 3. Ke3 Qa7+ 4. Ke2 Qa6+ 5. Ke3; 1. Bc1 Qxc1+ 2. Kf2 Qc8 3. Kg3 Kh7 4. Kf3 Qd8 5. Kg3 Kg7. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -4.72 from the side-to-move perspective.

SOLUTION 14 OF 24

Calculation test 14: Pawn fork

**VERIFIED LINE**

1. e6 Rg7+ 2. Kf2 Rde7 3. Ke3 Rxe6+ 4. Kd4 Rd7+ 5. Kc4 Rd8

WHY IT WORKS

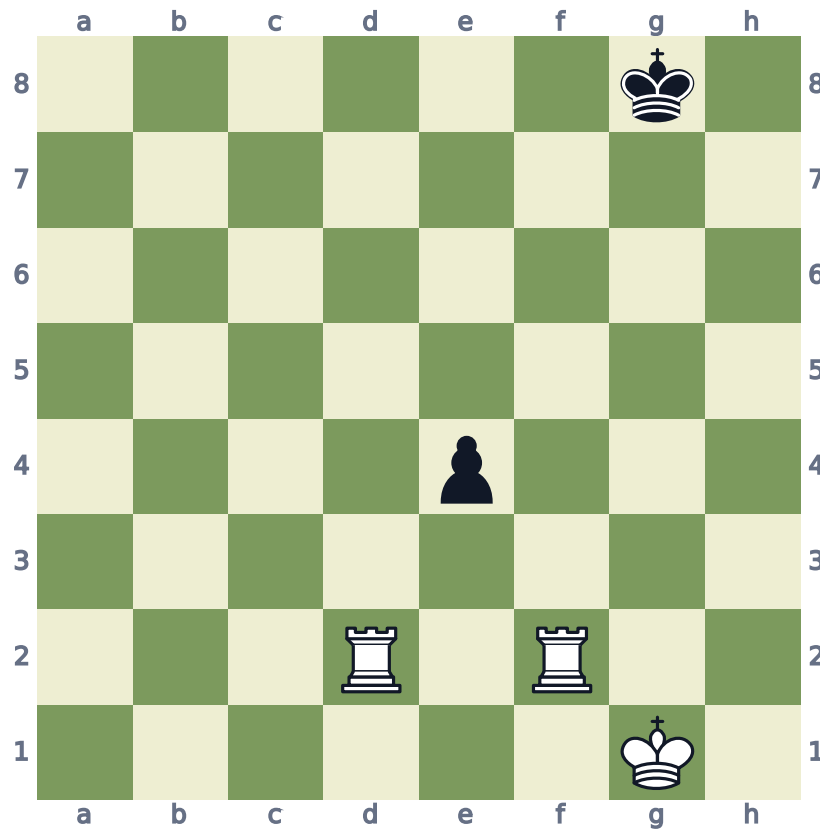
The best line is compared against these alternatives: 1. Kg2 Rde7 2. Kg3 Rxe5 3. Kg4 Rd5 4. Kh3 Rg7 5. Kh4 Rg1; 1. Kh2 Rde7 2. Kg3 Rxe5 3. Kg4 Rd5 4. Kh3 Rg7 5. Kh4 Rg1. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -6.85 from the side-to-move perspective.

SOLUTION 15 OF 24

Calculation test 15: Pawn fork - reverse view



VERIFIED LINE

1. e3 Rg2+ 2. Kf7 Rde2 3. Ke6 Rxe3+ 4. Kd5 Rd2+ 5. Kc4

WHY IT WORKS

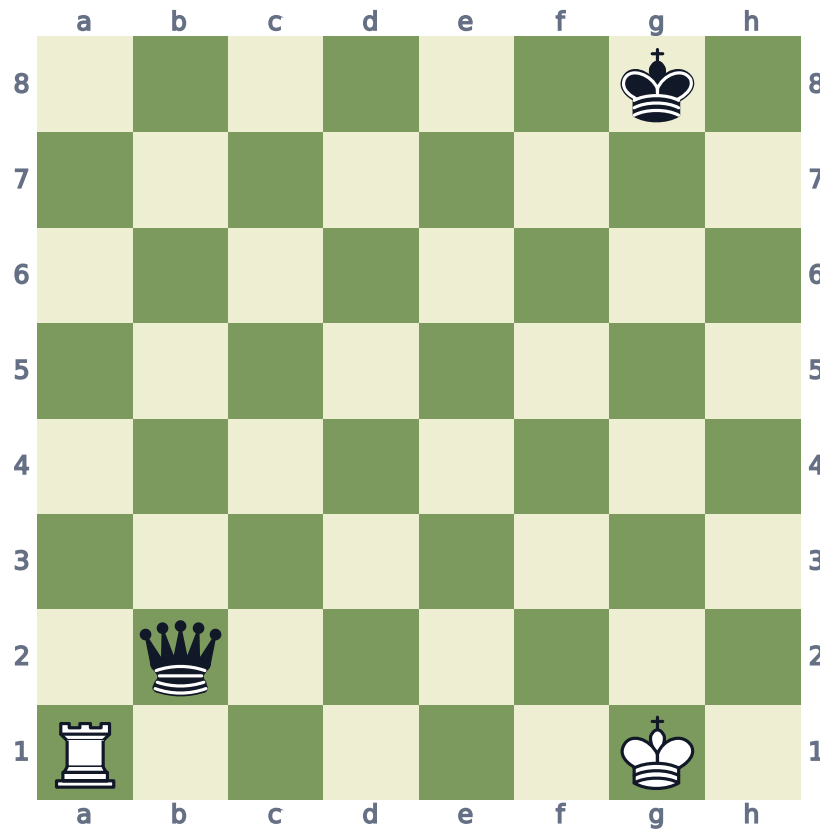
The best line is compared against these alternatives: 1. Kg7 Rde2 2. Kg6 Re3 3. Kh5 Rxe4 4. Kg5 Rd4 5. Kg6 Rd3; 1. Kh7 Rde2 2. Kh6 Rxe4 3. Kh5 Rd4 4. Kg5 Rd3 5. Kg4 Re3. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -7.12 from the side-to-move perspective.

SOLUTION 16 OF 24

Calculation test 16: Capture with check - reverse view



VERIFIED LINE

1. Qxa1+ Kf2 2. Qf6+ Ke3 3. Qc3+ Kf4 4. Qb2 Ke4

WHY IT WORKS

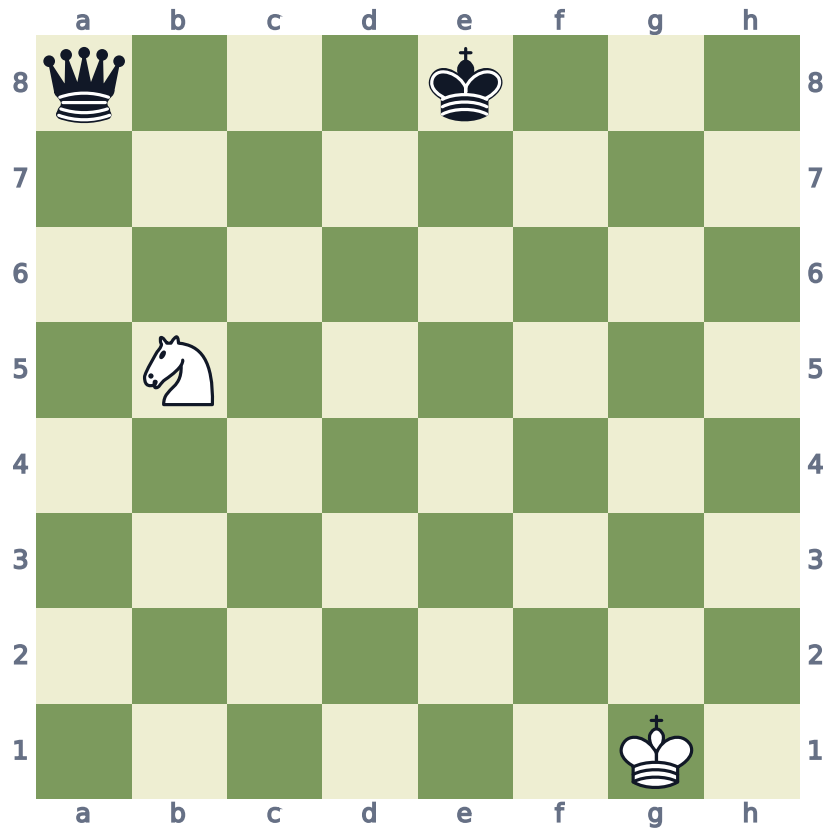
The best line is compared against these alternatives: 1. Qg7+ Kf2 2. Qxa1 Ke3 3. Qe5+ Kf3 4. Qd4 Ke2; 1. Qd4+ Kg2 2. Qxa1 Kg3 3. Qe5+ Kg4 4. Qe4+ Kg5 5. Qe3+ Kf5. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: +5.06 from the side-to-move perspective.

SOLUTION 17 OF 24

Calculation test 17: Royal fork



VERIFIED LINE

1. Nc7+ Kd7 2. Nxa8 Kd6 3. Nb6 Ke6 4. Nc4 Kd5 5. Ne3+ Ke4

WHY IT WORKS

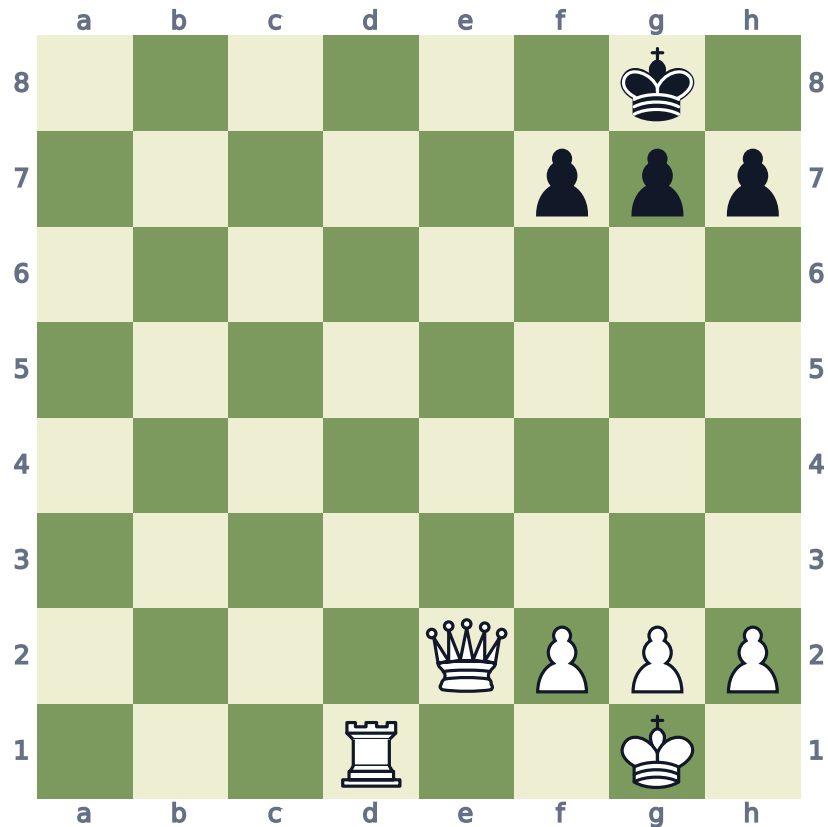
The best line is compared against these alternatives: 1. Nd4 Qa1+ 2. Kf2 Qxd4+ 3. Kf3 Qe5 4. Kg4 Qb8 5. Kf5 Ke7; 1. Kf2 Qa2+ 2. Ke3 Qb3+ 3. Ke4 Qxb5 4. Kd4 Qb4+ 5. Ke5 Kd7. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -0.05 from the side-to-move perspective.

SOLUTION 18 OF 24

Calculation test 18: Back-rank geometry



VERIFIED LINE

1. Qe8#

WHY IT WORKS

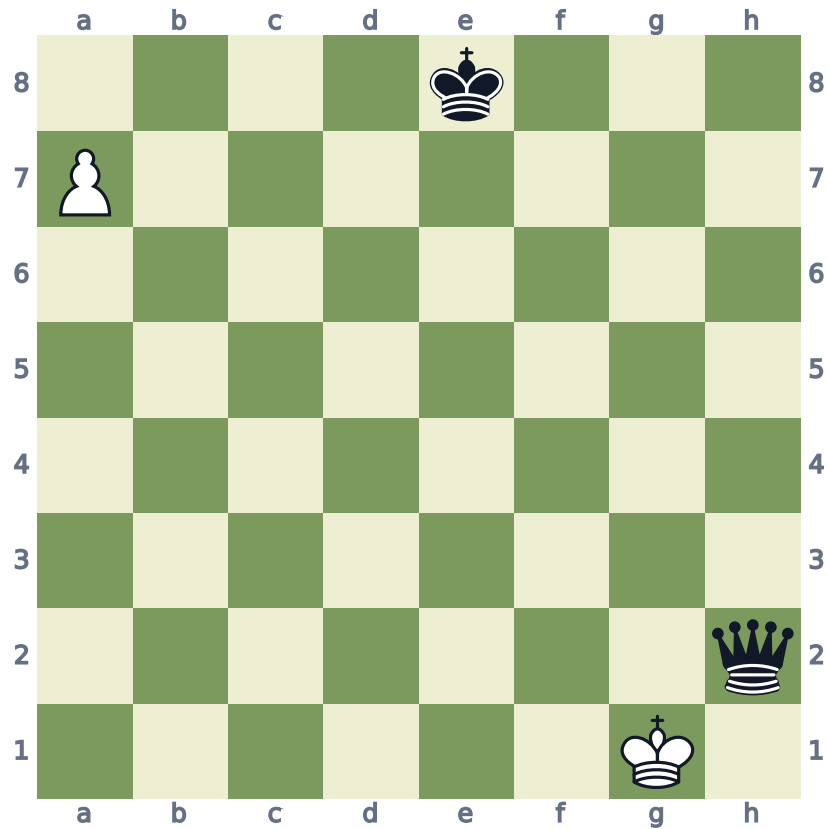
The best line is compared against these alternatives: 1. Rd8#; 1. f3 g5 2. Rd8+ Kg7 3. Qb2+ f6 4. Rd7+ Kg6 5. Qb1+ f5. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: mate in 1.

SOLUTION 19 OF 24

Calculation test 19: Promotion tactic



VERIFIED LINE

1. Kxh2 Ke7 2. Kh3 Kd6 3. a8=Q Ke5 4. Kh4 Kd4 5. Qa1+ Kd3

WHY IT WORKS

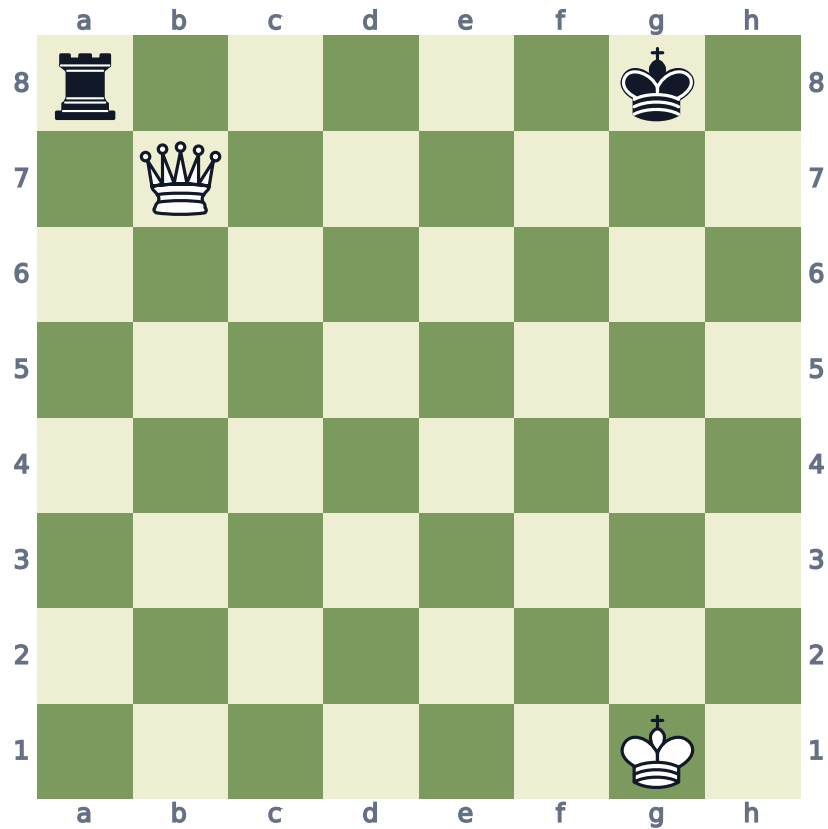
The best line is compared against these alternatives: 1. Kf1 Qa2 2. a8=Q+ Qxa8 3. Ke2 Ke7 4. Kd3 Kd6 5. Kd4 Qe8. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: +5.39 from the side-to-move perspective.

SOLUTION 20 OF 24

Calculation test 20: Capture with check



VERIFIED LINE

1. Qxa8+ Kf7 2. Qd5+ Kf6 3. Kg2 Kg7 4. Qe6 Kh7 5. Kf3 Kg7

WHY IT WORKS

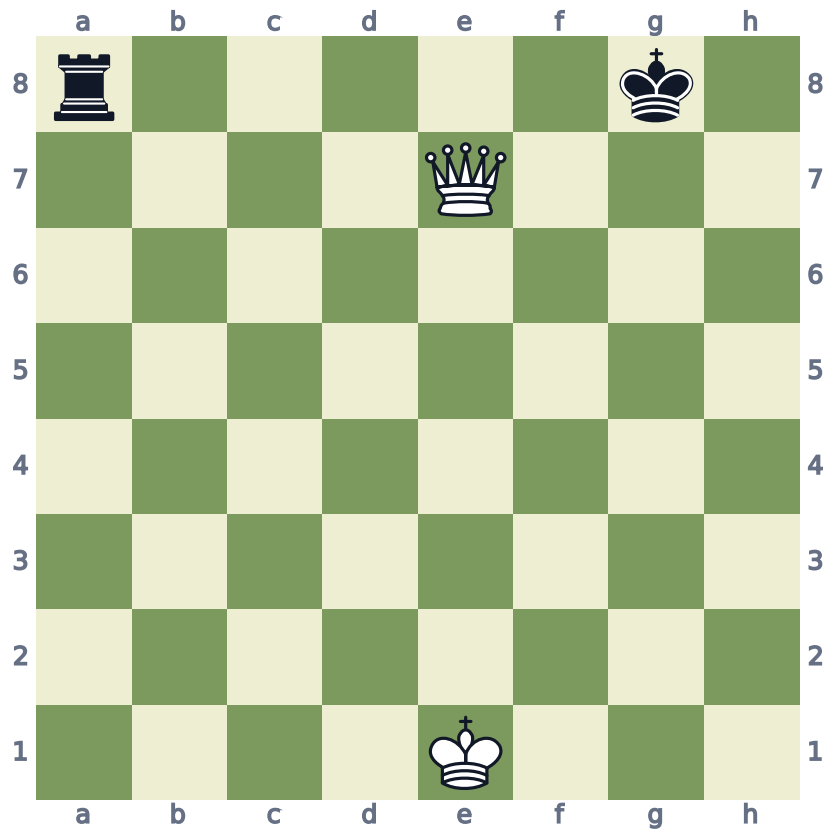
The best line is compared against these alternatives: 1. Qd5+ Kg7 2. Qg2+ Kf7 3. Qxa8 Ke6 4. Kh2 Ke5; 1. Qg2+ Kf7 2. Qxa8 Ke6 3. Qa5 Kd6 4. Kg2 Kc6 5. Kh3 Kd7. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: +5.16 from the side-to-move perspective.

SOLUTION 21 OF 24

Calculation test 21: Rook double attack - reverse view

**VERIFIED LINE**

1. Rf8 Ke2 2. Rf7 Qe6 3. Kf8 Qc8+ 4. Ke7 Qc5+ 5. Kf6 Qd4+

WHY IT WORKS

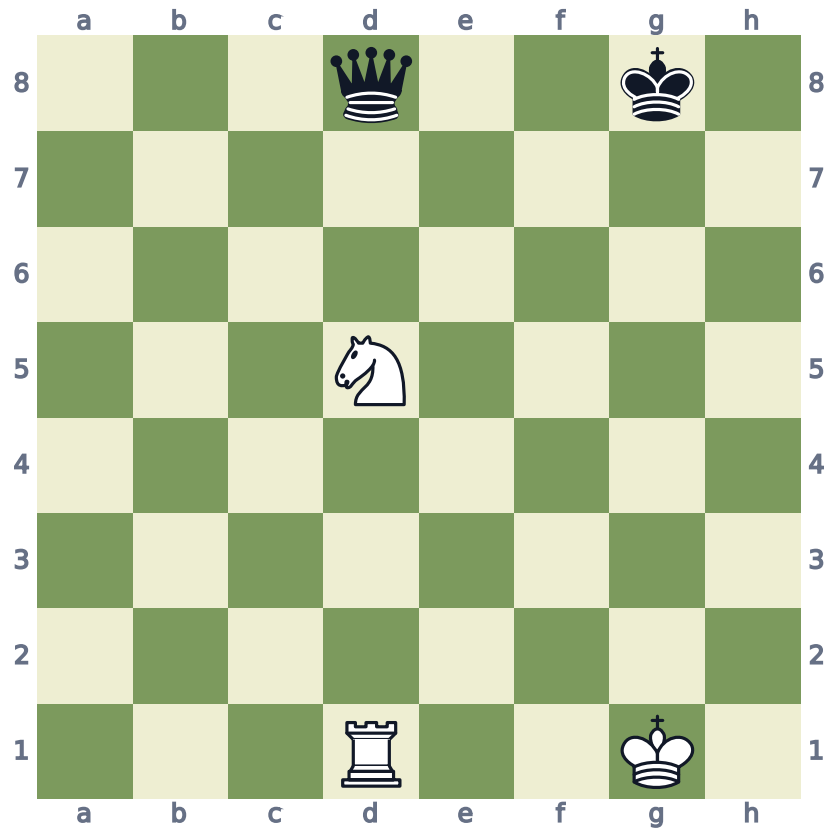
The best line is compared against these alternatives: 1. Ra1+ Kf2 2. Ra8 Qg5+ 3. Kh7 Qf5+ 4. Kh6 Qf6+ 5. Kh7 Qf7+; 1. Ra6 Ke2 2. Rg6 Kf3 3. Rg1 Qa7 4. Rg6 Kf4 5. Rg7 Qb8+. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -2.44 from the side-to-move perspective.

SOLUTION 22 OF 24

Calculation test 22: Discovered attack



VERIFIED LINE

1. Rd3 Qh4 2. Kf1 Kg7 3. Ne3 Qh1+ 4. Kf2 Qh5 5. Rd7+ Kg8

WHY IT WORKS

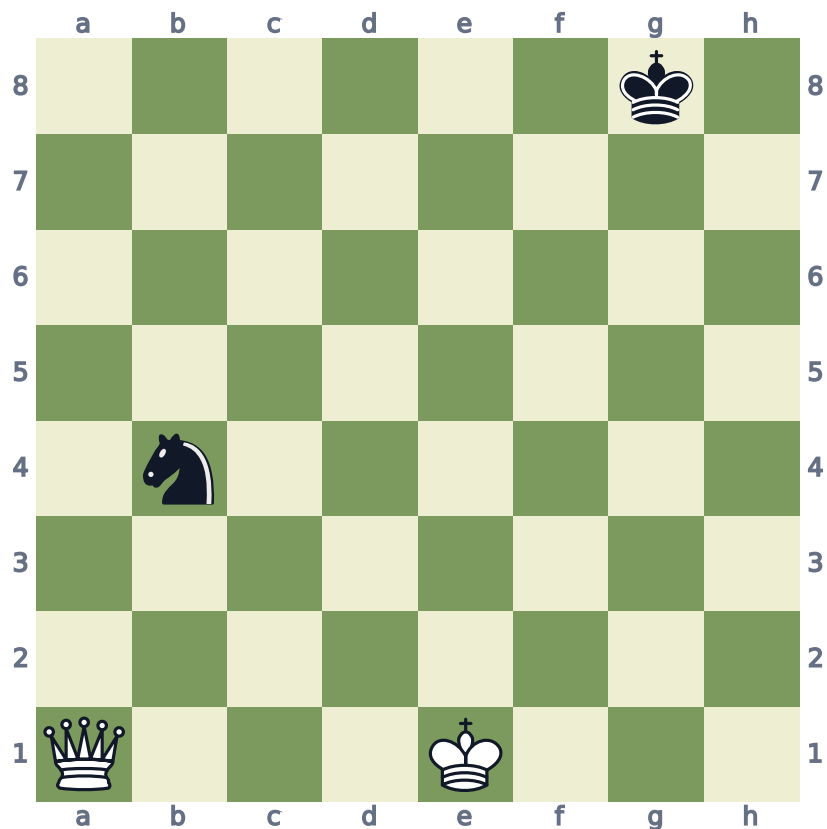
The best line is compared against these alternatives: 1. Kf1 Qg5 2. Rd3 Kf7 3. Rf3+ Ke8 4. Nf4 Qg4 5. Re3+ Kf7; 1. Ne3 Qf8 2. Rd2 Qh6 3. Rg2+ Kf7 4. Re2 Qg5+ 5. Kf2. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -0.06 from the side-to-move perspective.

SOLUTION 23 OF 24

Calculation test 23: Royal fork - reverse view



VERIFIED LINE

1. Nc2+ Kf2 2. Nxa1 Ke3 3. Nb3 Ke4 4. Nd2+ Ke5 5. Kf7 Kd5

WHY IT WORKS

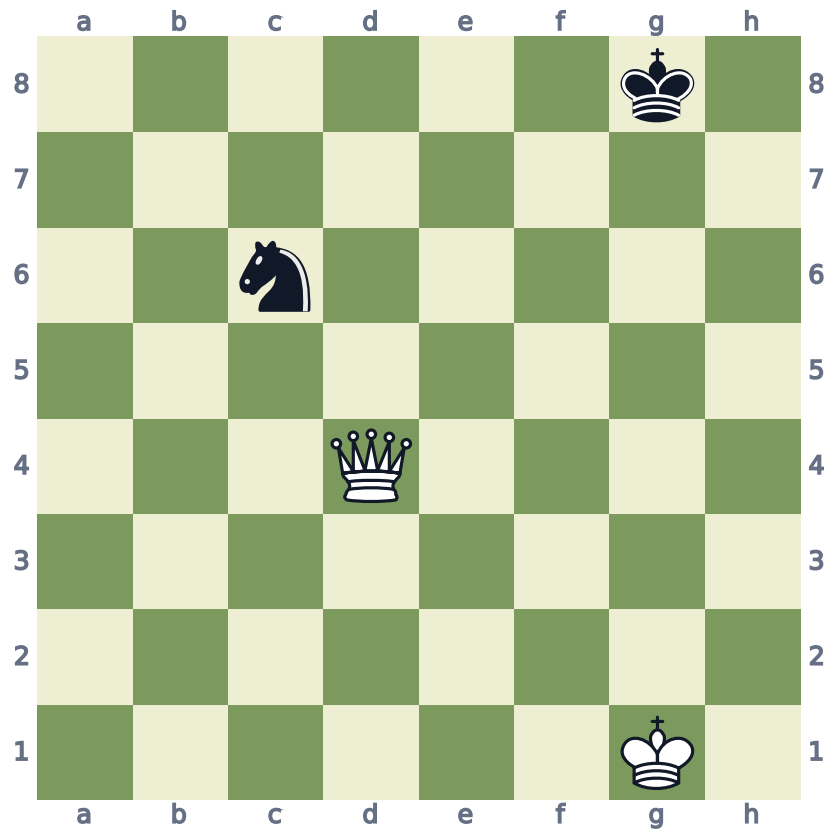
The best line is compared against these alternatives: 1. Kf7 Qa7+ 2. Ke6 Qb6+ 3. Kd5 Qxb4 4. Kc6 Kd1 5. Kd5 Kc1; 1. Nd5 Qa2 2. Kf7 Qxd5+ 3. Kf6 Kf2 4. Kg6. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: -0.06 from the side-to-move perspective.

SOLUTION 24 OF 24

Calculation test 24: Loose queen - reverse view



VERIFIED LINE

1. Nxd4 Kf1 2. Kf7 Ke1 3. Nb5 Kf2 4. Nd6

WHY IT WORKS

The best line is compared against these alternatives: 1. Kf7 Qc4+ 2. Kf6 Qxc6+ 3. Kf5 Qc2+ 4. Kf4 Qe2 5. Kf5; 1. Nb4 Qxb4 2. Kf7 Qe4 3. Kf6 Qe1 4. Kf5 Kh2 5. Kg5 Qa5+. The purpose is not only to find the move, but to maintain legal geometry and evaluate the final position.

SCORING

2 candidates, 3 main line, 2 best defense, 2 endpoint, 1 time discipline = 10 points Engine assessment: +0.02 from the side-to-move perspective.

REVIEW

Final mixed review

1

Select five missed positions and solve them in random order without motif names.

2

Explain why the best defense fails in each of those positions.

3

Reverse the board and describe whether the same geometry is still easy to recognize.

4

Find one example of the theme in your own game or a Fritz training game.

5

Retest under a shorter clock only after calculation is accurate without time pressure.

RETRIEVAL SCHEDULE

Repeat missed positions tomorrow, then after 3, 7, 14, and 30 days. Remove a position only after two correct solutions with a complete explanation.